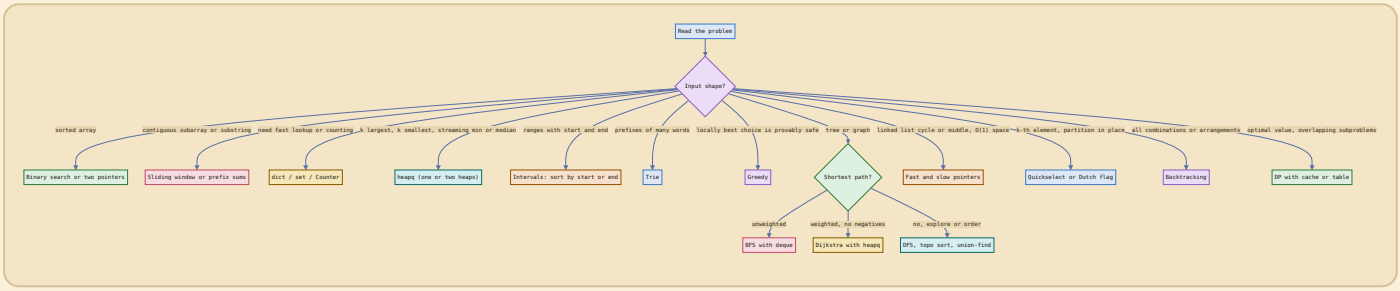
Chapter 19 — Python for Coding Interviews

Coding interviews (the "LeetCode-style" rounds at most tech companies — LeetCode is the best-known website of practice problems, and the online "judges" that run your code against hidden tests) test two things at once: whether you know the classic **data-structure and algorithm (DSA) patterns**, and whether you can express them cleanly in a language. Python is the most popular interview language because it is short: a heap (a structure that always gives you the smallest item fast, §19.5), a hash map (key → value lookup table, i.e. a `dict`, §19.3), a queue (first in, first out, §19.4) and a binary search (finding an item in sorted data by repeatedly halving the range, §19.6) are each one import away. But that shortness hides costs — `list.pop(0)` looks innocent and is O(n), `[[0]*n]*m` looks like a grid and is a bug.

This chapter is the Python toolkit for DSA. For each pattern you get the idea, the Python tools that implement it, a tiny runnable example with `assert` tests, its complexity (how its running time and memory grow with input size), and the mistakes people make. It assumes Chapters 5–7 (lists, dicts, functions) and 11 (iterators). All scripts are in `code/`; they were run with Python 3.14 (everything is 3.12-compatible) and the output shown is real.

What you'll learn

#	Topic	Script
19.1	Big-O refresher and the complexity cheat table for built-ins	`01_builtins_complexity.py`
19.2	Arrays & strings: two pointers, sliding window, prefix sums	`02_arrays_strings.py`
19.3	Hashing: `dict`, `set`, `Counter`, `defaultdict`	`03_hashing.py`
19.4	Stacks and queues: `list`, `deque`, monotonic stack/deque	`04_stacks_queues.py`
19.5	Heaps: `heapq`, the max-heap trick, top-k, merge k, two heaps (running median, maximise capital)	`05_heaps.py`
19.6	Sorting with keys; binary search by hand and with `bisect`; weighted random pick	`06_sorting_bisect.py`
19.7	Linked lists and trees as Python classes; serialize/deserialize a tree	`07_linked_lists_trees.py`
19.8	Graphs: adjacency lists, BFS/DFS, grids, Dijkstra, topological sort, recursion limit	`08_graphs.py`
19.9	Union-Find (disjoint sets), Kruskal	`09_union_find.py`
19.10	Dynamic programming: `@cache` (top-down) and tabulation (bottom-up)	`10_dynamic_programming.py`
19.11	Backtracking	`11_backtracking.py`
19.12	Bit tricks and math (`gcd`, `pow(b, e, m)`, `isqrt`, `comb`, sieve)	`12_bits_math.py`
19.13	Tries: prefix search, autocomplete, wildcard search, word search on a board	`13_tries.py`
19.14	Greedy: jump game, gas station, candy, interval scheduling	`14_greedy.py`
19.15	Intervals: merge, insert, intersect, meeting rooms (heap and sweep line)	`15_intervals.py`
19.16	Fast & slow pointers: cycle start, middle, palindrome list, happy number, find duplicate	`16_fast_slow_pointers.py`
19.17	Sort & search: merge sort on a list, quicksort, quickselect, Dutch national flag	`17_sort_search.py`
19.18	Math & geometry: spiral matrix, rotate, reverse a 32-bit int, collinear points, Josephus	`18_math_geometry.py`
19.19	Common pitfalls	`19_pitfalls.py`
19.20	How to talk through a solution	—



19.1 Big-O and the built-ins cheat table

What it is

Big-O notation describes how running time (or memory) grows with the input size n, ignoring constant factors — the fuel-consumption class of a car, not the litres for today's trip. O(1) does not grow with n, O(n) grows in proportion, O(n log n) is good sorting, O(n²) is "every pair". An **amortised** O(1) operation is occasionally expensive but averages out to constant time: `list.append` sometimes reallocates and copies, but rarely enough that the average stays O(1).

Why it matters

Interviewers always ask "what's the complexity?", and Python hides loops inside innocent-looking calls: `x in my_list` scans, `my_list.insert(0, x)` shifts every element, `s = s + ch` copies the string.

How it works — the cheat table

CPython's `list` is a dynamic array (a block of memory that is reallocated bigger when it fills up) of pointers (memory addresses of the actual objects); `dict` and `set` are hash tables (§19.3); `collections.deque` is a doubly-linked list (each block points to both the previous and the next) of fixed-size blocks; `heapq` works on a plain list arranged as a binary heap (§19.5). The costs follow from those designs (average case; hash tables can degrade to O(n) with pathological collisions — many keys deliberately or unluckily landing in the same slot):

Structure	Operation	Cost	Note
`list`	`a[i]`, `a[i] = x`, `len(a)`	O(1)	array indexing
	`a.append(x)`, `a.pop()`	O(1) amortised	end of the array
	`a.insert(i, x)`, `a.pop(i)`, `del a[i]`	O(n)	shifts the tail; `pop(0)` is the trap
	`x in a`, `a.index(x)`, `a.count(x)`, `min/max/sum`	O(n)	linear scan
	`a[i:j]`, `a.copy()`, `a + b`	O(k)	slicing **copies** k elements
	`a.sort()`, `sorted(a)`	O(n log n)	Timsort (Python's built-in sort), stable (equal items keep their order); O(n) if already sorted
	`a.reverse()`, `a[::-1]`	O(n)	
`deque`	`append`, `appendleft`, `pop`, `popleft`	O(1)	the right queue type
	`dq[i]` in the middle	O(n)	O(1) at either end
`dict`	`d[k]`, `d[k] = v`, `del d[k]`, `k in d`, `d.get`	O(1) avg	keys must be hashable (have a fixed hash — immutable types)
	iterate over keys/values/items	O(n)	insertion-ordered (3.7+)
`set`	`add`, `remove`, `discard`, `x in s`	O(1) avg	
	`s | t`, `s - t`	O(len s + len t) / O(len s)	
	`s & t`	O(min(len s, len t))	
`heapq`	`heappush`, `heappop`, `heappushpop`	O(log n)	min-heap on a list
	`h[0]` (peek)	O(1)	
	`heapify(list)`	O(n)	faster than n pushes
	`nlargest(k, it)`, `nsmallest(k, it)`	O(n log k)	
`bisect`	`bisect_left/right`	O(log n)	list must be sorted
	`insort`	O(n)	search is log n, the insert shifts
`str`	`s[i]`, `len(s)`	O(1)	immutable
	`s + t`, slicing, `s.replace`, `s.split`	O(n)	always builds a new string
	`"".join(parts)`	O(total length)	the right way to build strings
`Counter`	`Counter(iterable)`	O(n)	`most_common(k)` O(n log k)

Python `int`s have **arbitrary precision** (no overflow), so `(lo + hi) // 2` is safe — but arithmetic on huge numbers is not O(1); multiplying two d-digit numbers costs more than constant time. For interview-sized numbers, treat it as O(1).

The script measures the two most important rows:

```
def pop_front_times(n):
    def drain_list():
        items = list(range(n))
        while items:
            items.pop(0)        # O(n) each: every other element shifts left
    def drain_deque():
        items = deque(range(n))
        while items:
            items.popleft()     # O(1) each
    return timeit.timeit(drain_list, number=1), timeit.timeit(drain_deque, number=1)
```

Output (your machine's numbers will differ; the shape will not):

```
      n |   x in list |  x in set |  pop(0) all | popleft all
   1000 |      5.3 us |   0.03 us |     0.07 ms |     0.02 ms
   5000 |     26.3 us |   0.04 us |     1.06 ms |     0.13 ms
  20000 |    108.3 us |   0.03 us |    24.50 ms |     0.51 ms
  50000 |    321.3 us |   0.18 us |   174.89 ms |     1.47 ms
```

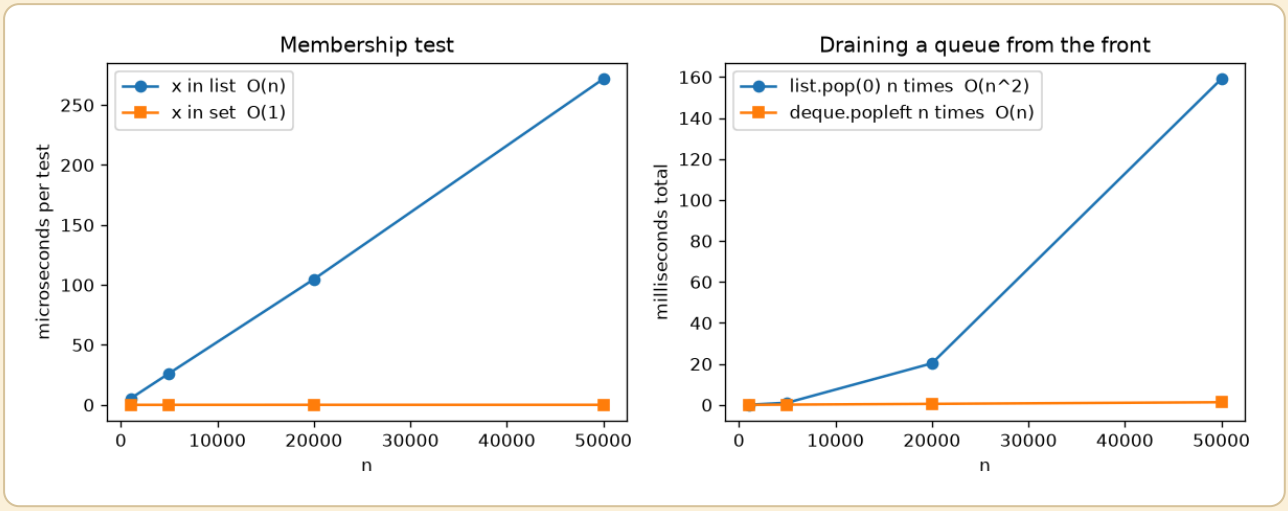


Figure — `x in list` grows linearly while `x in set` stays flat; draining a list from the front is quadratic, a deque is linear.

Common mistakes

- `x in list` inside a loop (O(n²) total) when a `set` makes it O(n); a list as a queue with `pop(0)`.
- Forgetting that slicing copies: `helper(nums[1:])` in recursion adds O(n) per call — pass indices instead.
- Building a string with `+=` in a loop (CPython sometimes optimises it in place, but you cannot rely on it) — collect pieces and `"".join` once.

In the interview

30-second answer: "Lists are dynamic arrays — O(1) index and append, O(n) insert/delete at the front and membership. Dicts and sets are hash tables with O(1) average lookup. For a queue I use `deque`, for a priority queue (always hands out the most urgent item next) `heapq`, and sorting is O(n log n) Timsort."

- *Why is `append` amortised O(1)?* → The array over-allocates; when full it grows by a proportion of its size and copies, so the total copying over n appends is O(n).

- *Worst case for a dict lookup?* → O(n) if many keys collide; in practice with good hashes it's O(1).
- *Space complexity of `sorted(a)`?* → O(n) for the new list (Timsort also uses up to n/2 temporary space).

19.2 Arrays and strings: two pointers, sliding window, prefix sums

These three patterns turn many "check every pair/subarray" (a subarray is a contiguous — unbroken — run of elements; a substring is the same for strings) O(n²) problems into O(n).

What it is

- **Two pointers** — two indices move through the data, usually from both ends toward each other (sorted pair sum, palindrome — text that reads the same backwards) or both forward at different speeds (a *slow* "write" pointer and a *fast* "read" pointer for in-place filtering — changing the existing list rather than building a new one).
- **Sliding window** — indices `left` and `right` bound a contiguous window, like a caterpillar: `right` grows it one step at a time, `left` shrinks it when a rule is broken. Each index only moves forward, so the total work is O(n).
- **Prefix sums** — precompute `prefix[k] = sum of the first k elements`. Then the sum of any range `nums[i:j]` is `prefix[j] − prefix[i]` in O(1).

How it works

Two pointers on a sorted array (pair with sum = target): start at both ends. If the sum is too small, only moving `lo` up can increase it; if too big, only moving `hi` down can decrease it. Each step discards one element forever → O(n).

Variable sliding window (longest substring with no repeated character): expand `right`, remembering each character's last index in a dict; if the new character was already seen inside the window, jump `left` past its earlier copy. The answer is the largest `right − left + 1`.

Prefix sums + hash map (count subarrays summing to k, negatives allowed): a subarray `(i, j]` sums to k exactly when `prefix[j] − prefix[i] = k`, i.e. `prefix[i] = prefix[j] − k`. Walk once, keep a count of prefix values seen so far, and add `seen[running − k]` at each step. Worked numbers: for `[1, 1, 1]` and k = 2, running sums are 1, 2, 3; at 2 we look up 0 (seen once, the empty prefix) → 1; at 3 we look up 1 (seen once) → 1; total 2.

```python
def two_sum_sorted(nums, target):
    lo, hi = 0, len(nums) - 1
    while lo < hi:
        s = nums[lo] + nums[hi]
        if s == target:
            return lo, hi
        if s < target:
            lo += 1          # need a bigger sum -> move the small end up
        else:
            hi -= 1          # need a smaller sum -> move the big end down
    return None

def longest_unique_substring(s):
    last_seen = {}
    left = best = 0
    for right, ch in enumerate(s):
        if ch in last_seen and last_seen[ch] >= left:
            left = last_seen[ch] + 1      # shrink: jump past the earlier copy
        last_seen[ch] = right
        best = max(best, right - left + 1)
    return best

def min_window_at_least(nums, target):   # positive numbers only
    left = total = 0
    best = float("inf")
    for right, x in enumerate(nums):
        total += x
        while total >= target:            # shrink while still valid
            best = min(best, right - left + 1)
            total -= nums[left]
            left += 1
    return 0 if best == float("inf") else int(best)

class RangeSum:
    def __init__(self, nums):
        self.prefix = [0, *accumulate(nums)]   # prefix[k] = sum(nums[:k])
    def query(self, i, j):                     # half-open [i, j)
        return self.prefix[j] - self.prefix[i]

def count_subarrays_sum_k(nums, k):
    seen = defaultdict(int)
    seen[0] = 1           # the empty prefix
    running = count = 0
    for x in nums:
        running += x
        count += seen[running - k]
        seen[running] += 1
    return count

assert two_sum_sorted([1, 3, 4, 6, 9], 10) == (0, 4)
assert longest_unique_substring("pwwkew") == 3
assert min_window_at_least([2, 3, 1, 2, 4, 3], 7) == 2
assert RangeSum([3, -1, 4, 1, 5]).query(1, 4) == 4
assert count_subarrays_sum_k([1, 1, 1], 2) == 2
```

("Half-open `[i, j)`" in `RangeSum` means i is included and j is not, like Python slices.) The script also has `is_palindrome` (two pointers from both ends, skipping non-alphanumerics — anything that isn't a letter or digit) and `remove_duplicates_sorted` (slow/fast pointers, in place). Output:

```
prefix: [0, 3, 2, 6, 7, 12]
two_sum_sorted([1,3,4,6,9], 10) -> (0, 4)
longest_unique_substring('pwwkew') -> 3
min_window_at_least([2,3,1,2,4,3], 7) -> 2
count_subarrays_sum_k([3,4,7,-2,2,1,4,2], 7) -> 6
all asserts passed
```

Python helpers worth knowing

Need	Python
index + value	`for i, x in enumerate(nums)`
adjacent pairs	`for a, b in itertools.pairwise(nums)` (3.10+) or `zip(nums, nums[1:])`
running totals	`itertools.accumulate(nums)`; `accumulate(nums, max)` for running max
reverse	`nums[::-1]` (copy) or `nums.reverse()` (in place)
letter → 0..25	`ord(ch) - ord("a")`; back with `chr(i + ord("a"))`
character class	`ch.isalnum()`, `ch.isdigit()`, `ch.isalpha()`, `ch.lower()`
strings are immutable	convert to `list(s)`, edit, then `"".join(chars)`

Trade-offs

Pattern	Needs	Time	Space
Two pointers (pair sum)	sorted input	O(n) (+ O(n log n) to sort)	O(1)
Hash map (pair sum, 19.3)	nothing	O(n)	O(n)
Sliding window	a condition that stays monotonic (only ever moves in one direction) as the window grows/shrinks (e.g. positive numbers)	O(n)	O(1)–O(alphabet)
Prefix sums	static array	O(n) build, O(1) query	O(n)

When not to use it

A shrinking window relies on "adding makes it bigger, removing makes it smaller". With **negative numbers** that breaks (shrinking may *increase* the sum) — use prefix sums + hash map instead, as in `count_subarrays_sum_k`. If the array changes between queries, prefix sums must be rebuilt; use a Fenwick/segment tree (tree structures that support both updates and range sums in O(log n)).

Common mistakes

- Off-by-one in window length (it is `right − left + 1`); forgetting `seen[0] = 1` (misses subarrays starting at index 0).
- Moving `left` backwards in the unique-substring problem — the `last_seen[ch] >= left` check prevents it.

In the interview

30-second answer: "When I see contiguous subarrays or substrings I think sliding window — expand right, shrink left while invalid, O(n). For pairs in a sorted array, two pointers from both

ends. For range sums or 'subarray sums to k', prefix sums with a hash map of seen prefixes."

- *Why is the window O(n) when it has a nested* `while`*?* → Each index enters and leaves the window at most once; the inner loop's total iterations across the whole run are ≤ n.
- *Two-sum on unsorted data?* → Hash map in O(n) time/O(n) space; sorting + two pointers is O(n log n) and loses original indices unless you sort `(value, index)` pairs.
- *What is* `prefix` *for* `[2, 4, 1]`*?* → `[0, 2, 6, 7]`.
- *Sum of* `nums[1:3]` *from it?* → `prefix[3] − prefix[1] = 5`.

19.3 Hashing: `dict`, `set`, `Counter`, `defaultdict`

What it is

A **hash table** stores items in buckets chosen by a **hash** of the key, giving O(1) average insert and lookup. Python's four flavours: `dict` (key → value), `set` (keys only), `collections.Counter` (a dict subclass that counts; missing keys read as 0) and `collections.defaultdict(factory)` (creates `list()`, `int()`, `set()`… the first time a missing key is accessed). Only **hashable** objects can be keys — `int`, `str`, `tuple` of hashables, `frozenset`; convert a list to a `tuple` first.

Why it matters

A large share of easy/medium problems are "trade O(n) memory for O(1) lookup": two-sum, anagrams (words made of the same letters, like "listen"/"silent"), duplicates, frequency counting, "have I seen this before?".

How it works

```
def two_sum(nums, target):
    index_of = {}
    for i, x in enumerate(nums):
        if target - x in index_of:        # the complement was seen earlier
            return index_of[target - x], i
        index_of[x] = i
    return None

def is_anagram(a, b):
    return Counter(a) == Counter(b)

def group_anagrams(words):
    groups = defaultdict(list)
    for w in words:
        groups[tuple(sorted(w))].append(w)   # key must be hashable -> tuple
    return list(groups.values())

def longest_consecutive(nums):
    values = set(nums)
    best = 0
    for x in values:
        if x - 1 not in values:           # x starts a run -> count it once
            length = 1
            while x + length in values:
                length += 1
            best = max(best, length)
    return best

assert two_sum([2, 7, 11, 15], 9) == (0, 1)
assert longest_consecutive([100, 4, 200, 1, 3, 2]) == 4

c = Counter("mississippi")
print("Counter:", c)
print("most_common(2):", c.most_common(2))
print("missing key:", c["z"])
print("c - Counter('sip'):", c - Counter("sip"))
```

Output:

```
Counter: Counter({'i': 4, 's': 4, 'p': 2, 'm': 1})
most_common(2): [('i', 4), ('s', 4)]
missing key: 0
c - Counter('sip'): Counter({'i': 3, 's': 3, 'm': 1, 'p': 1})
group_anagrams: [['eat', 'tea', 'ate'], ['tan', 'nat'], ['bat']]
all asserts passed
```

`longest_consecutive` looks like a nested loop but is O(n): the inner `while` only runs from the start of each run, so every number is visited by it at most once.

Handy API

Want	Code
read with default	`d.get(k, 0)`
count / group	`Counter`, `defaultdict(int)`, `defaultdict(list)` + `.append`
insert if missing, return value	`d.setdefault(k, [])`
top k counts	`Counter(xs).most_common(k)`
counter arithmetic	`c1 + c2`, `c1 - c2` (drops ≤ 0), `c1 & c2` (min), `c1 | c2` (max)
fixed alphabet count	`[0] * 26` indexed by `ord(ch) - 97` — also hashable as `tuple(...)`
set algebra	`a & b`, `a | b`, `a - b`, `a ^ b`

Trade-offs

Hashing costs O(n) extra memory and loses order-based tricks. If the input is sorted, two pointers use O(1) space. For keys drawn from a tiny fixed range (26 letters), an array is faster and simpler than a dict.

Common mistakes

- Using a list as a key (`TypeError: unhashable type: 'list'`) — use `tuple(lst)`.
- Reading a `defaultdict` with `d[k]` just to check it: that **inserts** the key. Use `k in d`.
- Mutating a dict while iterating over it (`RuntimeError: dictionary changed size during iteration`) — iterate over `list(d)`.

In the interview

30-second answer: "I use a dict when I need to remember something about values I've already seen — index, count, group — which turns an O(n²) pair search into O(n). Keys must be hashable, so I use tuples for composite keys. `Counter` and `defaultdict` remove boilerplate (repetitive set-up code)."

- *How would you make a dict key from a list of counts?* → `tuple(counts)`.
- *`Counter.most_common()` with no argument?* → All items sorted by count, O(n log n); with k it uses a heap, O(n log k).
- *Can a `set` be a dict key?* → No; a `frozenset` can.

19.4 Stacks and queues: `list` and `deque`

What it is

- A **stack** is last-in, first-out (LIFO), like a pile of plates: a `list` with `append` (push) and `pop()` — both O(1) at the end.
- A **queue** is first-in, first-out (FIFO), like a checkout line: `collections.deque` with `append` and `popleft` — both O(1). (`queue.Queue` is a slower thread-safe queue for concurrency, Chapter 16 — not what interviews want.)
- A **monotonic stack** keeps its elements in increasing (or decreasing) order by popping anything that breaks the order before pushing — "next greater/smaller element" in O(n). A **monotonic deque** also drops elements that leave a window from the front — sliding-window maximum in O(n).

How it works

Valid parentheses: push openers; on a closer, the top of the stack must be its matching opener; at the end the stack must be empty.

Next greater element: walk left to right holding indices whose values have not yet found a bigger element. When a new value `x` arrives, it *is* the answer for every smaller value on top of the stack — pop them and record `x`. Each index is pushed and popped once → O(n).

```
def valid_parentheses(s):
    pairs = {")": "(", "]": "[", "}": "{"}
    stack = []
    for ch in s:
        if ch in "([{":
            stack.append(ch)
        elif not stack or stack.pop() != pairs[ch]:
            return False
    return not stack

def next_greater(nums):
    result = [-1] * len(nums)
    stack = []                          # indices, values decreasing
    for i, x in enumerate(nums):
        while stack and nums[stack[-1]] < x:
            result[stack.pop()] = x
        stack.append(i)
    return result

def sliding_window_max(nums, k):
    dq = deque()                        # indices, values decreasing
    out = []
    for i, x in enumerate(nums):
        if dq and dq[0] <= i - k:
            dq.popleft()                # front fell out of the window
        while dq and nums[dq[-1]] <= x:
            dq.pop()                    # smaller values can never be the max
        dq.append(i)
        if i >= k - 1:
            out.append(nums[dq[0]])
    return out

assert valid_parentheses("([]{})") and not valid_parentheses("(]")
assert next_greater([2, 1, 2, 4, 3]) == [4, 2, 4, -1, -1]
assert sliding_window_max([1, 3, -1, -3, 5, 3, 6, 7], 3) == [3, 3, 5, 5, 6, 7]
```

The script also has `daily_temperatures` (same monotonic idea, storing distances), a `MinStack` (each entry stores `(value, min_so_far)` so `get_min` is O(1)), and `eval_rpn` for Reverse Polish Notation (writing operators after their operands, `3 4 +`, evaluated with a stack). Note the division there: `int(a / b)` truncates toward zero as the problem requires, while `a // b` floors (−7 // 2 = −4). Deque tour output:

```
deque: [0, 1, 2, 3, 4]
popleft -> 0 | pop -> 4 | now deque([1, 2, 3])
after rotate(1): deque([3, 1, 2])
deque(maxlen=3) after 0..5: deque([3, 4, 5], maxlen=3)
sliding_window_max: [3, 3, 5, 5, 6, 7]
all asserts passed
```

`deque(maxlen=k)` silently drops from the opposite end when full — a ready-made "last k items" buffer.

Common mistakes

- `stack.pop()` or `stack[-1]` on an empty list raises `IndexError` — check `if stack` first (the `not stack or ...` short-circuit above — `or` stops as soon as the left side is true, so `stack.pop()` never runs on an empty stack).
- Storing values instead of indices in a monotonic stack when you later need distances or window positions.

In the interview

> **30-second answer:** "Stack = list with append/pop. Queue = deque with append/popleft, never list.pop(0). When the question is 'next greater/smaller' or 'span', I use a monotonic stack of indices — O(n) because each index is pushed and popped once."

- *Implement a queue with two stacks?* → Push to `inbox`; to dequeue, if `outbox` is empty move everything from `inbox` to `outbox` (reversing it), then pop `outbox`. Amortised O(1).
- *Why does `sliding_window_max` pop smaller values from the back?* → A smaller value to the left of a bigger one can never be the maximum of any future window.
- *Which end does `deque.append` add to?* → Right. `appendleft` adds to the left.
- *What does `deque.rotate(1)` do?* → Moves the last element to the front.

19.5 Heaps with `heapq`

What it is

A **binary heap** is a complete binary tree (every level full except possibly the last, which fills from the left) stored in an array where every parent is ≤ its children (a **min-heap**), so the smallest item is at index 0. `heapq` provides functions that treat a plain `list` as a min-heap — there is no heap class. A heap is the standard **priority queue** (always serve the most urgent item next).

How it works

- Children of index i live at 2i+1 and 2i+2. `heappush` appends and "sifts up" (swaps the item with its parent while it is smaller); `heappop` moves the last item to the root and "sifts down" (swaps it with its smaller child while it is bigger) — both O(log n).
- **Max-heap trick:** `heapq` only does min-heaps, so push `-x` and negate on the way out. (Python 3.14 adds public `heapq.heappush_max` / `heappop_max` / `heapify_max`, but most interview platforms run older versions, so the negation trick is the portable answer.)
- **Tuples as entries:** `(priority, item)` compares by priority first. If priorities tie, Python compares the next field — add a unique counter `(priority, count, item)` when items themselves aren't comparable.
- **Top-k pattern:** to find the k largest of n items, keep a **min-heap of size k**; whenever it exceeds k, pop the smallest. The root is then the k-th largest. O(n log k) time, O(k) space.

```python
def kth_largest(nums, k):
    heap = []
    for x in nums:
        heapq.heappush(heap, x)
        if len(heap) > k:
            heapq.heappop(heap)       # drop the smallest
    return heap[0]

def top_k_frequent(words, k):
    counts = Counter(words)
    return heapq.nsmallest(k, counts, key=lambda w: (-counts[w], w))  # ties alphabetical

def merge_k_sorted(lists):
    heap = [(lst[0], i, 0) for i, lst in enumerate(lists) if lst]
    heapq.heapify(heap)
    out = []
    while heap:
        val, i, j = heapq.heappop(heap)       # i breaks ties
        out.append(val)
        if j + 1 < len(lists[i]):
            heapq.heappush(heap, (lists[i][j + 1], i, j + 1))
    return out

class RunningMedian:
    """low = max-heap (negated) of the smaller half, high = min-heap of the larger half."""
    def __init__(self):
        self.low, self.high = [], []
    def add(self, x):
        heapq.heappush(self.low, -x)
        heapq.heappush(self.high, -heapq.heappop(self.low))
        if len(self.high) > len(self.low):
            heapq.heappush(self.low, -heapq.heappop(self.high))
    def median(self):
        if len(self.low) > len(self.high):
            return float(-self.low[0])
        return (-self.low[0] + self.high[0]) / 2
```

Output of the tour in `__main__`:

```
heapified: [1, 2, 8, 3, 5] smallest: 1
pop order: [1, 2, 3, 5, 8]
max via negation: 8
priority queue first: (1, 'fix prod bug')
nlargest(2): [9, 7]
running medians: [5.0, 10.0, 5.0, 4.0]
max capital (k=2): 4
all asserts passed
```

Notice `heapified` is **not** sorted — only `h[0]` is guaranteed smallest. Popping repeatedly gives sorted order (that's heapsort, the sorting algorithm built on a heap).

The two-heaps pattern

Use **two heaps facing each other** when you must repeatedly split data into a "smaller" group and a "larger" group, and you need the boundary between them fast.

- **Running (streaming) median** — `RunningMedian` above. `low` is a max-heap holding the smaller half, `high` a min-heap holding the larger half. Two invariants: every value in `low` ≤ every value in `high`, and `len(low)` is equal to `len(high)` or one more. Then the median is `low`'s top (odd count) or the average of both tops (even count). `add` keeps both invariants by always passing the

new value through `low` into `high`, then moving one back if `high` got bigger. Trace for 5, 15, 1, 3:

add	`low` (max-heap)	`high` (min-heap)	median
5	{5}	{}	5
15	{5}	{15}	(5 + 15) / 2 = 10
1	{1, 5}	{15}	5
3	{1, 3}	{5, 15}	(3 + 5) / 2 = 4

- **Maximise capital** (`max_capital` in the script): you may do at most k projects; each needs `cost` capital to start and returns `profit`. One heap holds **locked** projects by cost (min-heap); the other holds **affordable** ones by profit (max-heap). Each round, move every project you can now afford from the first heap to the second, then take the most profitable. It is a greedy choice (always taking the best-looking option now, 19.14) that heaps make O(n log n).
- **Sliding-window median** is the hard follow-up: same two heaps, but values also leave the window. Heaps cannot delete from the middle cheaply, so you use **lazy deletion** — remember "remove x later" in a `Counter` and discard x when it reaches a top.

Trade-offs

Need	Best tool	Cost
k largest once, small k	`heapq.nlargest(k, xs)`	O(n log k)
k close to n	`sorted(xs)[-k:]`	O(n log n)
k-th element, average fast	quickselect (partition-based selection, 19.17)	O(n) average, $O(n^2)$ worst
repeated "give me the min" with inserts	heap	O(log n) each
delete arbitrary item	heap + "lazy deletion" (mark, skip when popped)	amortised

Common mistakes

- Assuming the heap list is sorted; forgetting to negate **both** on push and pop in a max-heap.
- Pushing `(priority, dict)` — on a tie, Python tries to compare dicts → `TypeError`. Add a tie-breaker counter.
- `heappush` onto a list that already held unordered items without `heapify`-ing it first.

In the interview

> **30-second answer:** "`heapq` is a min-heap on a list: push/pop O(log n), peek `h[0]` O(1), `heapify` O(n). For a max-heap I push negatives. For top-k I keep a size-k min-heap, O(n log k). Dijkstra, merge-k-lists and running median are all heap problems."

- *Why a min-heap for k largest?* → The root is the weakest of the current top k, so it's the one to evict.
- *Running median complexity?* → O(log n) per insert, O(1) per median.
- *Why not one sorted list with `insort`?* → The search is O(log n) but the insert shifts items, O(n) per add; two heaps are O(log n).

19.6 Sorting with keys; binary search and `bisect`

What it is

- `sorted(iterable, key=..., reverse=...)` returns a new list; `list.sort(...)` sorts in place and returns `None`. Both use **Timsort** — O(n log n) worst case, O(n) on already-sorted input, and **stable** (equal keys keep their original order).
- A **key function** maps each item to the value to compare (`key=len`, `key=str.lower`). Tuples compare element by element, so a tuple key gives multi-level sorting; negate a numeric field for "descending on this field only". `functools.cmp_to_key` adapts a pairwise comparator (a function `cmp(a, b)` returning negative, zero or positive) for the rare problems where order depends on the *pair* (e.g. "largest number").
- **Binary search** halves a sorted range each step: O(log n). `bisect_left(a, x)` = first index with `a[i] >= x`; `bisect_right(a, x)` = first index with `a[i] > x`; `insort` inserts keeping order. Since 3.10 they accept `key=`.

How it works

Binary search on the answer is the pattern interviewers love: when a yes/no test `ok(v)` is *monotonic* (false, …, false, true, …, true), binary-search the smallest v that is true. Example: the minimum eating speed to finish banana piles in h hours — if speed s works, any faster speed works too.

```
people = [("ana", 31), ("bo", 25), ("cy", 31), ("di", 25)]
print(sorted(people, key=lambda p: (p[1], p[0])))      # age, then name
print(sorted(people, key=lambda p: (-p[1], p[0])))     # age desc, name asc

def merge_intervals(intervals):
    out = []
    for start, end in sorted(intervals):
        if out and start <= out[-1][1]:
            out[-1][1] = max(out[-1][1], end)
        else:
            out.append([start, end])
    return out

def lower_bound(nums, target):        # same as bisect_left
    lo, hi = 0, len(nums)              # half-open [lo, hi)
    while lo < hi:
        mid = (lo + hi) // 2
        if nums[mid] < target:
            lo = mid + 1
        else:
            hi = mid
    return lo

def min_eating_speed(piles, hours):
    def ok(speed):
        return sum(ceil(p / speed) for p in piles) <= hours
    lo, hi = 1, max(piles)
    while lo < hi:
        mid = (lo + hi) // 2
        if ok(mid):
            hi = mid          # mid works -> try smaller
        else:
            lo = mid + 1
    return lo

a = [1, 3, 3, 3, 7, 9]
assert lower_bound(a, 3) == bisect.bisect_left(a, 3) == 1
assert bisect.bisect_right(a, 3) == 4
assert min_eating_speed([3, 6, 7, 11], 8) == 4
```

Output:

```
by age then name: [('bo', 25), ('di', 25), ('ana', 31), ('cy', 31)]
age desc, name asc: [('ana', 31), ('cy', 31), ('bo', 25), ('di', 25)]
itemgetter(1) (stable): [('bo', 25), ('di', 25), ('ana', 31), ('cy', 31)]
case-insensitive: ['Apple', 'banana', 'cherry']
by length desc: ['banana', 'cherry', 'Apple']
count of 3s via bisect: 3
insort 5: [1, 3, 3, 3, 5, 7, 9]
first age>=31 at index 2
weighted pick shares (expect ~0.17, 0.50, 0.33): [0.17, 0.5, 0.33]
all asserts passed
```

The script also contains `largest_number` (uses `cmp_to_key`: put `a` before `b` if `a+b > b+a` as strings) and `search_rotated` (binary search in a rotated sorted array: one half is always sorted, check whether the target lies in it).

Weighted random selection

Problem: given weights `w = [1, 3, 2]`, return index i with probability `w[i] / sum(w)` (1/6, 3/6, 2/6). Used for load balancing (spreading requests across several servers — see System Design Fundamentals) across servers of different sizes and for sampling.

Idea: lay the weights end to end on a number line using **prefix sums**: `[1, 4, 6]` means index 0 owns [0, 1), index 1 owns [1, 4), index 2 owns [4, 6). Draw a random integer r in [0, 6) and find which slot it falls in — the first prefix value **greater than** r, which is exactly `bisect_right`. Building is O(n), each pick O(log n).

```
class WeightedPicker:
    def __init__(self, weights, seed=None):
        self.prefix = list(accumulate(weights))     # [1, 3, 2] -> [1, 4, 6]
        self.rng = random.Random(seed)
    def pick(self):
        r = self.rng.randrange(self.prefix[-1])     # integer in [0, total)
        return bisect.bisect_right(self.prefix, r)
```

60,000 seeded picks gave shares `[0.17, 0.5, 0.33]` (output above). Python also has `random.choices(range(n), weights=w)`, which does the same prefix-sum + bisect internally — mention it, but interviewers usually want the hand-written version. Using `bisect_left` here is an off-by-one: r = 1 would map to index 0.

The two binary-search templates

Template	Loop	Update	Returns
Exact match	`while lo <= hi` on `[lo, hi]`	`lo = mid + 1` / `hi = mid - 1`	index or −1
Boundary (first true)	`while lo < hi` on `[lo, hi)`	`lo = mid + 1` / `hi = mid`	first position where condition holds

Pick one template and use it every time; most binary-search bugs come from mixing them (infinite loops when `lo = mid`).

Common mistakes

- `nums = nums.sort()` sets `nums` to `None`; `reverse=True` reverses the whole key, not one field.
- `bisect` on an unsorted list (no error, just wrong answers); thinking `insort` is O(log n) — the search is, the insertion is O(n).

In the interview

30-second answer: "Python sorts with Timsort, stable, O(n log n). I pass a `key` — tuples for multi-level ordering and negation for descending fields. For binary search I use `bisect_left/right` or a half-open `lo < hi` template, and for 'minimum value that satisfies X' I binary-search the answer space when X is monotonic."

- *Why does stability matter?* → You can sort by a secondary key first, then by the primary; equal primaries keep the secondary order.
- *Is there a TreeMap/sorted set in Python?* → Not built in. Use a sorted list + `bisect` (O(n) inserts), a heap, or the third-party `sortedcontainers.SortedList` (some platforms, such as LeetCode, preinstall it).
- *`bisect_left([1,2,2,3], 2)` and `bisect_right`?* → 1 and 3.
- *Sort strings by length, ties alphabetical?* → `sorted(ws, key=lambda w: (len(w), w))`.

19.7 Linked lists and trees in Python

What it is

Python has no built-in linked list or tree for interviews; you define tiny node classes. A **singly linked list** node has `val` and `next`; a **binary tree** node has `val`, `left`, `right`. A **binary search tree (BST)** additionally keeps every left-subtree (the node's left child plus everything below it) value < node < every right-subtree value. `@dataclass` (Chapter 10) writes the `__init__` for you; `from __future__ import annotations` lets a class mention itself in type hints.

How it works — linked lists

- **Dummy head:** start with a fake node so you never special-case "the list is empty" when building or merging.
- **Reversal:** walk with `prev`; point each node back at `prev`. Python's tuple assignment evaluates the whole right side first, so `head.next, prev, head = prev, head, head.next` is a correct one-liner.
- **Fast/slow pointers (Floyd, after the computer scientist Robert Floyd):** `slow` moves 1, `fast` moves 2. When `fast` reaches the end, `slow` is at the middle; if there is a cycle, they eventually meet. Compare nodes with `is` (identity).

```
@dataclass
class ListNode:
    val: int
    next: ListNode | None = None

def reverse(head):
    prev = None
    while head:
        head.next, prev, head = prev, head, head.next
    return prev

def has_cycle(head):
    slow = fast = head
    while fast and fast.next:
        slow, fast = slow.next, fast.next.next
        if slow is fast:
            return True
    return False

def merge_two(a, b):
    dummy = tail = ListNode(0)
    while a and b:
        if a.val <= b.val:
            tail.next, a = a, a.next
        else:
            tail.next, b = b, b.next
        tail = tail.next
    tail.next = a or b
    return dummy.next
```

How it works — trees

- **DFS traversals:** pre-order (node, left, right), in-order (left, node, right — gives sorted order for a BST), post-order (left, right, node). Recursive versions are 3 lines; iterative versions use an explicit stack.
- **BFS / level order:** a `deque`; process `len(q)` nodes per outer loop to separate levels.

- **Validate a BST:** pass down the allowed `(lo, hi)` range — checking only parent vs child is a classic bug.
- **Lowest common ancestor (LCA — the deepest node that has both target nodes below it):** recurse; if both sides return something, the current node is the LCA.

```
@dataclass
class TreeNode:
    val: int
    left: TreeNode | None = None
    right: TreeNode | None = None

def inorder_iterative(root):
    out, stack, node = [], [], root
    while stack or node:
        while node:                 # go left as far as possible
            stack.append(node)
            node = node.left
        node = stack.pop()
        out.append(node.val)
        node = node.right
    return out

def level_order(root):
    if not root:
        return []
    levels, q = [], deque([root])
    while q:
        levels.append([])
        for _ in range(len(q)):
            node = q.popleft()
            levels[-1].append(node.val)
            q.extend(c for c in (node.left, node.right) if c)
    return levels

def max_depth(root):
    return 0 if not root else 1 + max(max_depth(root.left), max_depth(root.right))

def is_valid_bst(root, lo=float("-inf"), hi=float("inf")):
    if not root:
        return True
    if not lo < root.val < hi:
        return False
    return is_valid_bst(root.left, lo, root.val) and is_valid_bst(root.right, root.val, hi)

def lca(root, p, q):
    if root is None or root.val in (p, q):
        return root
    left, right = lca(root.left, p, q), lca(root.right, p, q)
    return root if left and right else left or right
```

`build_tree([3, 9, 20, None, None, 15, 7])` in the script turns LeetCode's level-order list format into nodes so you can test locally. Output:

```
reversed list: [5, 4, 3, 2, 1]
inorder: [9, 3, 15, 20, 7] preorder: [3, 9, 20, 15, 7]
levels: [[3], [9, 20], [15, 7]] depth: 3
serialized: 3,9,#,#,20,15,#,#,7,#,#
all asserts passed
```

Serialize and deserialize a binary tree

Serialize = turn a tree into a string (to store or send it); **deserialize** = rebuild the identical tree. A plain pre-order list of values is not enough — `[1, 2]` could be 2 as a left or a right child. The fix is to also write a marker (`#`) for every missing child. Then pre-order is unambiguous: read a value, build the node, and the next tokens are *exactly* its left subtree followed by its right subtree.

```
def serialize(root):
    out, stack = [], [root]
    while stack:                        # iterative pre-order
        node = stack.pop()
        if node is None:
            out.append("#"); continue
        out.append(str(node.val))
        stack.append(node.right)        # right first, so left is popped first
        stack.append(node.left)
    return ",".join(out)

def deserialize(data):
    tokens = iter(data.split(","))      # a shared iterator = "read the next token"
    def build():
        tok = next(tokens)
        if tok == "#":
            return None
        node = TreeNode(int(tok))
        node.left = build()
        node.right = build()
        return node
    return build()
```

Both are O(n) time and space; for a tree with n nodes the string has 2n + 1 tokens (n values + n + 1 `#`). The script checks the round trip `deserialize(serialize(t)) == t`. The recursive `build` has the usual depth limit — for very deep trees use the level-order format that `build_tree` reads instead.

Complexity

All traversals are O(n) time. Space is O(h) for DFS (h = tree height: log n if balanced — both sides of every node roughly equal in size — n if a degenerate "linked-list" tree) and O(w) for BFS (w = widest level, up to n/2).

Common mistakes

- A mutable default like `def f(node, path=[])` in tree recursion (19.19).
- `if node.val:` when the value can be `0` — test `if node:` / `is None`.
- Recursing on a 10^5-deep degenerate tree → `RecursionError`; use the iterative version.

In the interview

30-second answer: "I define small `ListNode` / `TreeNode` classes. For lists: dummy head, fast/slow pointers, in-place reversal with three pointers. For trees: recursive DFS for clarity, an explicit stack if depth can be large, and BFS with a deque for level order. Space is O(height)."

- *Detect where a cycle starts?* → After slow and fast meet, move one pointer to the head; advance both by 1; they meet at the cycle start.
- *Inorder successor in a BST?* → If there's a right child, the leftmost node of it; else the nearest ancestor you reached via a left edge.

19.8 Graphs: adjacency lists, BFS, DFS, Dijkstra, topological sort

What it is

A **graph** is a set of **nodes** (vertices) joined by **edges**, which may be **directed** (one-way) and **weighted** (have a cost). The standard Python representation is an **adjacency list**: a `defaultdict(list)` mapping node → neighbours (or `(neighbour, weight)` pairs). A **grid** is an implicit graph whose cells neighbour the 4 (or 8) adjacent cells.

- **BFS (breadth-first search)** visits nodes in rings of increasing distance using a queue → shortest paths in **unweighted** graphs.
- **DFS (depth-first search)** goes as deep as possible before backtracking (recursion or a stack) → connectivity, cycles, topological order, flood fill.
- **Dijkstra's algorithm** finds shortest paths with **non-negative** weights using a min-heap of `(distance, node)`.
- **Topological sort** orders a **DAG** (directed acyclic graph) so every edge goes from earlier to later — course prerequisites, build order.

How it works

BFS: put the start in a deque and in `visited`; repeatedly pop the front and push unvisited neighbours, **marking them visited when enqueued** (marking on pop lets the same node be enqueued many times).

Dijkstra (lazy version):

1. `dist = {src: 0}`, heap = `[(0, src)]`.
2. Pop the smallest `(d, u)`. If `d` is larger than the best known `dist[u]`, it's a stale entry — skip it.
3. For each edge `u → v` with weight w: if `d + w < dist[v]`, update and push `(d + w, v)`.
4. Complexity O((V + E) log V).

Topological sort, Kahn's algorithm: count in-degrees (the number of edges pointing *into* each node); start with all nodes of in-degree 0; pop one, append it to the order, decrement its neighbours' in-degrees and enqueue any that reach 0. If the order has fewer than V nodes, there is a cycle.

Topological sort, DFS: three colours — white (unvisited), grey (on the current path), black (finished). Meeting a grey node means a back edge (an edge pointing back to a node still on the current path) = cycle. Append each node when it finishes (post-order) and reverse at the end.

```python
def bfs_shortest_path(g, start, goal):
    parent = {start: None}          # doubles as the visited set
    q = deque([start])
    while q:
        node = q.popleft()
        if node == goal:
            path = []
            while node is not None:
                path.append(node)
                node = parent[node]
            return path[::-1]
        for nxt in g.get(node, []):
            if nxt not in parent:   # mark when ENQUEUED
                parent[nxt] = node
                q.append(nxt)
    return None

def dijkstra(g, src):
    dist = {src: 0}
    heap = [(0, src)]
    while heap:
        d, u = heapq.heappop(heap)
        if d > dist.get(u, float("inf")):
            continue                    # stale entry
        for v, w in g.get(u, []):
            nd = d + w
            if nd < dist.get(v, float("inf")):
                dist[v] = nd
                heapq.heappush(heap, (nd, v))
    return dist

def topo_sort_kahn(n, prereqs):         # (a, b): b before a
    g = defaultdict(list)
    indeg = [0] * n
    for a, b in prereqs:
        g[b].append(a)
        indeg[a] += 1
    q = deque(i for i in range(n) if indeg[i] == 0)
    order = []
    while q:
        u = q.popleft()
        order.append(u)
        for v in g[u]:
            indeg[v] -= 1
            if indeg[v] == 0:
                q.append(v)
    return order if len(order) == n else None   # None -> cycle
```

The script also has `dfs_recursive`, `dfs_iterative` (pushes neighbours in reverse so the order matches the recursive one), `num_islands` (grid flood fill with a stack and the four direction offsets), and `topo_sort_dfs`. Output:

```
adjacency: {'A': ['B', 'C'], 'B': ['A', 'D'], 'C': ['A', 'D'], 'D': ['B', 'C', 'E'], 'E': ['D']}
dfs recursive: ['A', 'B', 'D', 'C', 'E'] | iterative: ['A', 'B', 'D', 'C', 'E']
dijkstra from A: {'A': 0, 'B': 3, 'C': 1, 'D': 4}
course order (Kahn): [0, 1, 2, 3] | DFS: [0, 2, 1, 3]
recursion limit: 1000
depth 5000 -> RecursionError
after setrecursionlimit(10_000): depth 5000
all asserts passed
```

Both topological orders are valid — a DAG usually has several.

The recursion limit

CPython refuses to nest calls deeper than `sys.getrecursionlimit()` (1000 by default) and raises `RecursionError`. Python has **no tail-call optimisation** (some languages reuse the current stack frame when a function's very last action is calling itself; Python doesn't), so even tail-recursive code counts. Options:

1. Prefer the **iterative** version (explicit stack/deque) for graphs/trees that may be deep — safest.
2. `sys.setrecursionlimit(10**6)` — accepted on many judges, but very deep recursion can still crash the interpreter by exhausting the real C stack (the memory the interpreter itself uses for nested calls) in some situations, so treat it as a workaround, not a fix.

Grid template

```
for dr, dc in ((1, 0), (-1, 0), (0, 1), (0, -1)):
    nr, nc = r + dr, c + dc
    if 0 <= nr < rows and 0 <= nc < cols and grid[nr][nc] == "1" and (nr, nc) not in seen:
        ...
```

Python's chained comparison `0 <= nr < rows` keeps bounds checks readable.

Trade-offs

Problem	Algorithm	Complexity
Shortest path, unweighted	BFS	O(V + E)
Shortest path, weights 0/1	0-1 BFS (deque, `appendleft` for 0-edges)	O(V + E)
Shortest path, non-negative weights	Dijkstra	O((V + E) log V)
Negative weights	Bellman–Ford (relaxes every edge V−1 times)	O(V·E)
All pairs, small V	Floyd–Warshall (DP over intermediate nodes)	O(V³)
Order with dependencies	Topological sort	O(V + E)
Connected components (groups of nodes reachable from each other), dynamic unions	Union-Find (19.9)	~O(α(n)) per op (effectively constant, see 19.9)

Common mistakes

- Marking visited on pop in BFS (duplicates in the queue, sometimes TLE — "time limit exceeded", the judge's verdict when code is too slow); a list instead of a set for `visited`.
- Using Dijkstra with negative edges (wrong answers).
- Forgetting isolated nodes that never appear in an edge list (loop over `range(n)`, not just over `g`).

In the interview

30-second answer: "I store graphs as `defaultdict(list)` adjacency lists. Unweighted shortest path → BFS with a deque, marking visited on enqueue. Weighted non-negative → Dijkstra with heapq, skipping stale entries. Dependencies → topological sort with Kahn's algorithm, which also detects cycles. For deep graphs I use iterative DFS because of Python's recursion limit."

- *Why doesn't Dijkstra work with negative edges?* → It finalises a node when popped assuming no later path can be shorter; a negative edge can break that.
- *Detect a cycle in an undirected graph?* → DFS that ignores the edge back to the parent, or union-find (an edge joining two nodes already in the same set).

19.9 Union-Find (disjoint set union)

What it is

Union-Find (a.k.a. **DSU**, disjoint set union) keeps track of which items belong to the same group while groups merge. It supports two operations: `find(x)` → the representative ("root") of x's group, and `union(a, b)` → merge the two groups. Analogy: companies merging — to know whether two employees work for the same firm, follow "who owns whom" up to the top parent company.

How it works

1. `parent[i] = i` initially (everyone is their own root).
2. `find` follows parents to the root. **Path compression** then points every node on that path directly at the root, flattening the tree.
3. `union` finds both roots; if different, attach the **smaller** tree under the larger (**union by size**) and decrement the component count. Return `False` if already joined — that is how you detect a cycle-closing edge.
4. With both optimisations, each operation is O(α(n)) amortised, where α is the inverse Ackermann function — ≤ 4 for any realistic n, so "effectively constant".

```python
class DSU:
    def __init__(self, n):
        self.parent = list(range(n))
        self.size = [1] * n
        self.components = n

    def find(self, x):
        root = x
        while self.parent[root] != root:
            root = self.parent[root]
        while self.parent[x] != root:          # path compression (iterative)
            self.parent[x], x = root, self.parent[x]
        return root

    def union(self, a, b):
        ra, rb = self.find(a), self.find(b)
        if ra == rb:
            return False
        if self.size[ra] < self.size[rb]:
            ra, rb = rb, ra
        self.parent[rb] = ra
        self.size[ra] += self.size[rb]
        self.components -= 1
        return True

def kruskal_mst(n, edges):              # edges (weight, u, v)
    dsu, total = DSU(n), 0
    for w, u, v in sorted(edges):
        if dsu.union(u, v):
            total += w
    return total

assert kruskal_mst(4, [(1, 0, 1), (4, 0, 2), (3, 1, 2), (2, 2, 3), (5, 1, 3)]) == 6
```

Output: `parent: [0, 0, 0, 3, 3, 5] components: 3` then `all asserts passed`. The script also solves "number of connected components" and "redundant connection" (the first edge whose `union` returns `False`). The iterative `find` avoids the recursion limit; in `self.parent[x], x = root, self.parent[x]` the right side (the old parent) is evaluated before `x` is rebound.

When not to use it

Union-Find cannot **split** groups and cannot give you paths — use BFS/DFS for those. For a one-off "count components" on a static graph, BFS/DFS is equally good.

In the interview

30-second answer: "Union-Find tracks groups under merging. `find` with path compression and `union` by size give near-constant amortised time. I use it for connected components, cycle detection in undirected graphs, Kruskal's MST (minimum spanning tree — the cheapest set of edges that connects every node), and 'accounts merge'-type problems."

- *Why union by size?* → Keeps trees shallow (height O(log n) even without compression).
- *Kruskal's complexity?* → O(E log E) for the sort; the DSU part is near-linear.

19.10 Dynamic programming: `@cache` and tabulation

What it is

Dynamic programming (DP) combines answers to smaller **overlapping subproblems**, computing each once. It applies when the problem has **optimal substructure** (the best answer is built from best answers to subproblems).

- **Top-down (memoisation — caching each result by its arguments):** write the natural recursion, then cache it with `functools.cache` (3.9+) or `lru_cache(maxsize=None)`.
- **Bottom-up (tabulation):** fill an array from the smallest subproblems upward with loops; no recursion, and you can often keep only the last row.

How it works — the recipe

1. **State:** what parameters describe a subproblem? (`i` = index, `(i, j)` = prefixes of two strings, `(i, capacity)`…)
2. **Transition:** how does a state's answer come from smaller states?
3. **Base cases.**
4. **Order** (bottom-up) or **memoise** (top-down).
5. **Answer location** (`dp[n]`, `dp[m][n]`, `max(dp)`…).

Worked example — coin change, coins `[1, 2, 5]`, amount 11: `dp[a] = 1 + min(dp[a − c])` over coins c ≤ a, `dp[0] = 0`. `dp[5] = 1` (one 5), `dp[10] = 2`, `dp[11] = 1 + min(dp[10], dp[9], dp[6]) = 1 + 2 = 3`.

```python
from functools import cache

@cache
def climb(n):                    # 1 or 2 steps at a time
    return 1 if n <= 1 else climb(n - 1) + climb(n - 2)

def coin_change(coins, amount):
    INF = amount + 1
    dp = [0] + [INF] * amount
    for a in range(1, amount + 1):
        for c in coins:
            if c <= a:
                dp[a] = min(dp[a], dp[a - c] + 1)
    return dp[amount] if dp[amount] != INF else -1

def lcs(a, b):
    m, n = len(a), len(b)
    dp = [[0] * (n + 1) for _ in range(m + 1)]    # NOT [[0]*(n+1)]*(m+1)
    for i in range(1, m + 1):
        for j in range(1, n + 1):
            if a[i - 1] == b[j - 1]:
                dp[i][j] = dp[i - 1][j - 1] + 1
            else:
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])
    return dp[m][n]

def edit_distance(a, b):
    @cache
    def go(i, j):
        if i == len(a):
            return len(b) - j
        if j == len(b):
            return len(a) - i
        if a[i] == b[j]:
            return go(i + 1, j + 1)
        return 1 + min(go(i + 1, j), go(i, j + 1), go(i + 1, j + 1))
    return go(0, 0)

def knapsack_01(weights, values, cap):
    dp = [0] * (cap + 1)
    for w, v in zip(weights, values):
        for c in range(cap, w - 1, -1):    # backwards: each item used once
            dp[c] = max(dp[c], dp[c - w] + v)
    return dp[cap]
```

Output:

```
climb.cache_info(): CacheInfo(hits=8, misses=11, maxsize=None, currsize=11)
coin_change([1,2,5], 11) = 3
edit_distance('horse','ros') = 3
knapsack = 9
all asserts passed
```

`cache_info()` shows only 11 real computations for `climb(10)` — without the cache the naive recursion makes 177 calls, and the count grows exponentially with n.

The script also includes `house_robber` (O(1)-space DP with two variables), `unique_paths` (one-row grid DP) and `lis_length` — the O(n log n) longest increasing subsequence that keeps `tails[k]` = smallest tail of an increasing run of length k+1 and uses `bisect_left` to place each number.

Common DP families

Family	State	Examples
1-D sequence	`dp[i]`	climbing stairs, house robber, decode ways, LIS (longest increasing subsequence)
Unbounded choice	`dp[amount]`	coin change, combination sum count
0/1 knapsack (pick items, each at most once, to maximise value within a weight limit)	`dp[capacity]` iterated backwards	subset sum, partition equal subset
Two strings	`dp[i][j]`	LCS (longest common subsequence), edit distance (fewest single-letter edits to turn one string into another), regex/wildcard matching
Grid	`dp[r][c]`	unique paths, minimum path sum
Intervals	`dp[i][j]` over lengths	burst balloons, palindrome partitioning

Top-down vs bottom-up

	`@cache` recursion	Tabulation
Writing it	closest to the recurrence (the formula expressing a state in terms of smaller states); quick	must choose iteration order
Computes	only reachable states	all states
Limits	recursion limit; function-call overhead	none
Memory tricks	hard	easy to keep one row (O(n) instead of O(n·m))

Common mistakes

- `@cache` on a function taking a **list** → `TypeError: unhashable type`. Pass tuples or indices.
- A module-level `@cache` function reused across test cases with different global data returns stale answers — define the cached helper **inside** the solving function (as `edit_distance` does) or call `.cache_clear()`.
- Building 2-D tables with `[[0] * n] * m` (all rows are the same list — see 19.19).
- Iterating 0/1 knapsack capacity **forwards** (turns it into unbounded knapsack).

In the interview

> **30-second answer:** "I first write the recurrence and state, then memoise it with `functools.cache` — that's top-down. If recursion depth or memory is a concern I convert to a bottom-up table and often compress it to one row. Complexity is number of states × work per state."

- *Complexity of edit distance?* → O(m·n) states × O(1) = O(m·n) time; O(m·n) space top-down, O(min(m, n)) with rolling rows.
- *`lru_cache` vs `cache`?* → `cache` is exactly `lru_cache(maxsize=None)`: unbounded, and cheaper than a bounded LRU because it never tracks recency; `lru_cache(maxsize=k)` evicts least-recently-used entries.
- *How to see how many states `@cache` stored?* → `f.cache_info().currsize`.

19.11 Backtracking

What it is

Backtracking builds candidate solutions step by step and abandons ("prunes") a partial candidate as soon as it cannot lead to a valid answer. It is DFS over a tree of choices. Template: **choose** → **explore** → **un-choose**.

How it works

1. Keep a mutable `path` (current partial solution) and maybe a `used` array/sets for constraints.
2. At each level, loop over the available choices; skip invalid ones (prune).
3. Append the choice, recurse, then pop it to restore the state.
4. When the path is complete, record **a copy** (`path[:]`).

```
def subsets(nums):
    out, path = [], []
    def go(start):
        out.append(path[:])              # copy! path keeps changing
        for i in range(start, len(nums)):
            path.append(nums[i])         # choose
            go(i + 1)                    # explore
            path.pop()                   # un-choose
    go(0)
    return out

def combination_sum(cands, target):      # reuse allowed
    cands = sorted(cands)
    out, path = [], []
    def go(start, remaining):
        if remaining == 0:
            out.append(path[:])
            return
        for i in range(start, len(cands)):
            if cands[i] > remaining:
                break                    # prune: later ones are bigger
            path.append(cands[i])
            go(i, remaining - cands[i])  # i, not i+1: reuse allowed
            path.pop()
    go(0, target)
    return out

def n_queens(n):
    cols, diag, anti = set(), set(), set()
    def go(r):
        if r == n:
            return 1
        count = 0
        for c in range(n):
            if c in cols or r - c in diag or r + c in anti:
                continue
            cols.add(c); diag.add(r - c); anti.add(r + c)
            count += go(r + 1)
            cols.remove(c); diag.remove(r - c); anti.remove(r + c)
        return count
    return go(0)
```

Output:

```
subsets: [[], [1], [1, 2], [1, 2, 3], [1, 3], [2], [2, 3], [3]]
combination_sum([2,3,6,7], 7): [[2, 2, 3], [7]]
8-queens solutions: 92
all asserts passed
```

The tests check `subsets`/`permute` against `itertools.combinations`/`permutations`, and `word_search` marks grid cells visited by overwriting them with `"#"` and restoring afterwards.

`itertools` shortcuts

When you just need to *enumerate*, the standard library already does it (Chapter 11): `permutations(xs, r)`, `combinations(xs, r)`, `combinations_with_replacement`, `product(xs, repeat=k)`. Interviewers usually want the hand-written version when pruning matters, but mentioning these shows fluency.

Complexity

Exponential by nature: subsets $O(2^n \cdot n)$, permutations $O(n! \cdot n)$. Pruning shrinks the real work but not the worst case.

Common mistakes

- Appending `path` instead of `path[:]` — every stored result becomes the same (final, usually empty) list.
- Forgetting to undo the choice (sets/`used` flags drift).
- Duplicates in input (e.g. subsets II): sort first and skip `nums[i] == nums[i-1]` when `i > start`.

In the interview

> **30-second answer:** "Backtracking is DFS over choices: choose, recurse, undo. I store a copy of the path at the leaves and prune early — sort candidates so I can break, and use sets for O(1) constraint checks, like columns and diagonals in N-Queens."

- *Why `r - c` and `r + c` for diagonals?* → Cells on the same "" diagonal share r − c; on the same "/" diagonal they share r + c.
- *How many subsets does a 4-element set have?* → $2^4 = 16$.

19.12 Bit tricks and math

What it is

Python `int`s are arbitrary-precision and behave like infinite **two's-complement** numbers (the standard binary encoding of negative integers, where −x is "flip all bits, then add 1") for bitwise operators: `&` (and), `|` (or), `^` (xor — exclusive or: 1 when exactly one bit is 1), `~` (not: `~x == -x - 1`), `<<`, `>>` (arithmetic shift — shifting while keeping the sign; floors for negatives). The `math` module and built-in `pow` cover interview number theory (properties of whole numbers: divisibility, primes, remainders).

Bit cheat table (x = 180 = 0b10110100, from the real output)

Trick	Expression	Result
test bit i	`x >> i & 1`	bit 2 → 1
set bit i	`x | (1 << i)`	set bit 0 → `0b10110101`
clear bit i	`x & ~(1 << i)`	
toggle bit i	`x ^ (1 << i)`	
lowest set bit	`x & -x`	4
clear lowest set bit	`x & (x - 1)`	`0b10110000`
power of two?	`x > 0 and x & (x - 1) == 0`	
count 1-bits	`x.bit_count()` (3.10+) or `bin(x).count("1")`	4
bits needed	`x.bit_length()`	8
format	`f"{x:08b}"`, `bin(x)`, `int("1011", 2)`	

XOR identities power "single number": `a ^ a = 0`, `a ^ 0 = a`, order doesn't matter. A bitmask (an integer whose bits are used as a row of on/off flags) `mask` from `0` to `2ⁿ − 1` enumerates all subsets (bit i set ⇒ item i chosen) — the basis of bitmask DP.

Math toolkit

Need	Python
gcd / lcm	`math.gcd(a, b)`, `math.lcm(a, b)` (3.9+, both accept many args)
modular power	`pow(b, e, m)` — fast square-and-multiply, O(log e)
modular inverse (the x with a·x ≡ 1 mod m; m prime or gcd = 1)	`pow(a, -1, m)` (3.8+)
integer square root	`math.isqrt(n)` — exact, no float error
combinations	`math.comb(n, k)`, `math.perm(n, k)`, `math.factorial(n)`
floor/ceil division	`a // b`, `-(-a // b)` or `math.ceil(a / b)` (float — careful with huge ints)
quotient and remainder	`divmod(a, b)`
infinity	`float("inf")`, `math.inf`; `sys.maxsize` is just a big int
big modulus	`MOD = 10**9 + 7`

Output of the script (excerpt):

```
~5 = -6  (two's complement: -(5+1))
-7 >> 1 = -4  (arithmetic shift floors)
gcd(84, 36) = 12 | lcm(4, 6) = 12
pow(2, 100, MOD) = 976371285
comb(5,2) = 10 | perm(5,2) = 20 | factorial(5) = 120
isqrt(10**20 + 1) = 10000000000
primes <= 30: [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]
2**100 = 1267650600228229401496703205376 (arbitrary precision)
divmod(-7, 2) = (-4, 1) | math.fmod(-7, 2) = -1.0
```

The script also writes `fast_pow` by hand (square-and-multiply: square the base each step and multiply it into the result when the current exponent bit is 1) and a Sieve of Eratosthenes (the classic way to list all primes up to n: cross out multiples starting at i^2, O(n log log n)).

Common mistakes

- Expecting 32-bit overflow: problems that say "return the 32-bit result" need masking, e.g. `x & 0xFFFFFFFF` and converting back if the top bit is set.
- `~x` is not "flip bits of an unsigned number"; it is `-x - 1`.
- Using `math.sqrt` on big integers (float precision) — use `math.isqrt`.
- `(a ** b) % m` builds a giant number first — use `pow(a, b, m)`.

In the interview

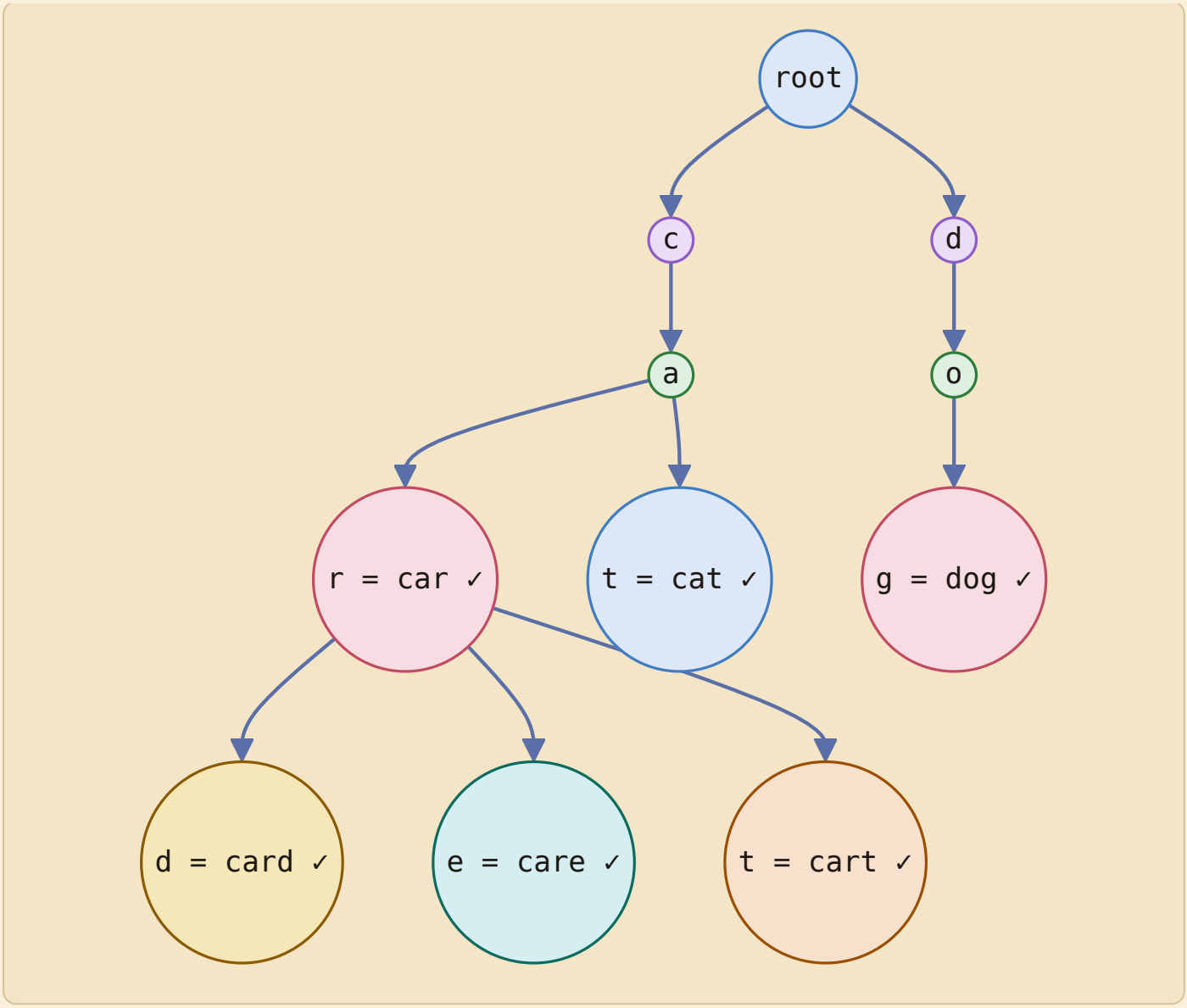
30-second answer: "Python ints never overflow, so I use `pow(b, e, m)` for modular exponentiation, `math.gcd`, `math.isqrt` and `math.comb` instead of hand-rolling. For bits: `x & (x−1)` drops the lowest set bit, `x & -x` isolates it, XOR cancels pairs."

- *Check power of four?* → Power of two and the single bit sits at an even position: `n > 0 and n & (n−1) == 0 and n & 0x55555555 != 0` (32-bit range), or replace the mask test with `n.bit_length() % 2 == 1` for any size. (In Python `&` binds tighter than `==`, unlike C, so no extra parentheses are needed.)
- *`6 & 5`, `6 | 5`, `6 ^ 5`?* → 4, 7, 3.
- *Modular inverse of 3 mod 7?* → `pow(3, -1, 7) = 5` (3·5 = 15 ≡ 1).

19.13 Tries (prefix trees)

What it is

A **trie** (say "try", from re*trie*val) is a tree that stores strings letter by letter. Each node represents a **prefix** (the first few letters of some word); each edge adds one letter; a flag on the node says "a whole word ends here". All words that start with "car" share the path c → a → r, so a trie answers **"which words start with this prefix?"** by walking the prefix once — no scanning of the whole word list.



The problem it solves

With a `set` of words, "is `car` a word?" is O(L) (L = word length, for hashing), but "does any word start with `ca`?" means checking every word — O(n·L). A trie makes both O(L), and it lets a search **stop early** the moment a path is not a prefix of anything, which is what makes word search on a board fast.

How it works

In Python a node is just a `dict` of children plus a field for the word that ends there. `setdefault(ch, TrieNode())` walks or creates in one step.

```python
class TrieNode:
    __slots__ = ("children", "word")        # many small nodes -> save memory
    def __init__(self):
        self.children = {}                   # letter -> TrieNode
        self.word = None                     # full word if one ends here

class Trie:
    def __init__(self):
        self.root = TrieNode()
    def insert(self, word):
        node = self.root
        for ch in word:
            node = node.children.setdefault(ch, TrieNode())
        node.word = word
    def _walk(self, s):
        node = self.root
        for ch in s:
            node = node.children.get(ch)
            if node is None:
                return None
        return node
    def search(self, word):
        node = self._walk(word)
        return node is not None and node.word is not None
    def starts_with(self, prefix):
        return self._walk(prefix) is not None
```

Problem 1 — autocomplete. Walk to the prefix node, then DFS below it, visiting children in alphabetical order, and collect the first `limit` words (`words_with_prefix` in the script).

Problem 2 — search with a wildcard. `.` matches any single letter. At a normal letter follow one child; at a `.` try **every** child (`any(...)` stops at the first success):

```python
def search_wildcard(self, pattern):
    def dfs(node, i):
        if i == len(pattern):
            return node.word is not None
        if pattern[i] == ".":
            return any(dfs(child, i + 1) for child in node.children.values())
        child = node.children.get(pattern[i])
        return child is not None and dfs(child, i + 1)
    return dfs(self.root, 0)
```

Problem 3 — word search on a board. Given a letter grid and a list of words, find every word you can trace through horizontally/vertically adjacent cells without reusing a cell. Running a separate grid search per word repeats work; instead put all words in one trie and run **one backtracking DFS from each cell that walks the trie at the same time**. A path that leaves the trie is abandoned at once.

```python
def find_words(board, words):
    trie = Trie()
    for w in words:
        trie.insert(w)
    rows, cols, found = len(board), len(board[0]), []

    def dfs(r, c, parent):
        ch = board[r][c]
        node = parent.children.get(ch)
        if node is None:
            return                          # no word continues this way
        if node.word is not None:
            found.append(node.word)
            node.word = None                # report each word once
        board[r][c] = "#"                   # mark visited in place
        for dr, dc in ((1, 0), (-1, 0), (0, 1), (0, -1)):
            nr, nc = r + dr, c + dc
            if 0 <= nr < rows and 0 <= nc < cols and board[nr][nc] != "#":
                dfs(nr, nc, node)
        board[r][c] = ch                    # undo (backtracking, 19.11)
        if not node.children and node.word is None:
            del parent.children[ch]         # prune a finished branch

    for r in range(rows):
        for c in range(cols):
            dfs(r, c, trie.root)
    return sorted(found)
```

Output of `13_tries.py`:

```
autocomplete 'car': ['car', 'card', 'care']
wildcard 'ca.e': True | '.o': False
words on board: ['eat', 'oat', 'oath']
all asserts passed
```

Complexity

Operation	Time	Notes
insert / search / starts_with	O(L)	L = length of the word or prefix
wildcard search	O(L) with no dots; up to O(26^d · L) with d dots	every dot branches
build for n words	O(total letters)	space: up to one node per letter
word search on an r × c board, max word length L	O(r·c·4·3^(L−1)) worst case	pruning makes it far faster in practice

When not to use it

- Only exact lookups ("is this a word?") → a `set` is simpler and uses less memory.
- Only one prefix query on a static list → `sorted` + `bisect` on the prefix is enough.

Common mistakes

- Treating "the path exists" as "the word exists" — `search("ca")` must be `False` when only "car" was inserted; that is what the end-of-word field is for.

- Word search: forgetting to restore the cell after the DFS, or reporting the same word twice when two paths spell it (clear `node.word` after the first find).
- Using `defaultdict(TrieNode)`-style auto-creation in `search` — a *read* then creates nodes and the trie grows on every miss.

In the interview

> **30-second answer:** "A trie stores words letter by letter; each node is a prefix, so insert, search and prefix checks are O(L). In Python a node is a dict of children plus an end-of-word marker. For a wildcard I branch into every child; for word search on a board I put all words in one trie and backtrack from every cell while walking the trie, pruning as soon as the path isn't a prefix."

- *Why use a trie instead of a hash set of all prefixes?* → A set of all prefixes costs O(total L^2) memory and cannot enumerate "words under this prefix"; a trie shares common prefixes.
- *How to return the top 3 suggestions by popularity?* → Store at each node the top-3 words for its prefix (updated on insert), making a lookup O(L).
- *Memory is too high?* → Use a **radix tree** (compress chains of single-child nodes into one edge), or an array of 26 children instead of a dict for lowercase-only input.

19.14 Greedy

What it is

A **greedy algorithm** builds the answer one step at a time and at each step takes the choice that looks best **right now**, never going back. It is correct only when you can argue that the local best choice never blocks the global best — typically with an **exchange argument**: "take any optimal answer; swapping its first choice for the greedy choice keeps it optimal."

The problem it solves

Many optimisation problems (find the best — max or min — answer) look like they need DP (19.10) over every combination. When a greedy rule is provably safe, the answer drops to a single pass, O(n) or O(n log n) after sorting, with O(1) extra space.

How it works

There is no single template; look for "keep only the best thing seen so far" or "sort, then take whatever still fits". Four classics:

Problem 1 — jump game. `nums[i]` is the maximum jump length from index i. *Can you reach the last index?* Track `farthest`, the furthest index reachable so far; if you ever stand beyond it, you are stuck. *Fewest jumps?* Think of BFS levels (19.8) without a queue: all indices reachable with j jumps form a range; when you reach the end of the current range you must jump once more, and the new range ends at the farthest point seen.

```python
def can_jump(nums):
    farthest = 0
    for i, n in enumerate(nums):
        if i > farthest:
            return False
        farthest = max(farthest, i + n)
    return True

def min_jumps(nums):
    jumps = end = farthest = 0
    for i in range(len(nums) - 1):
        farthest = max(farthest, i + nums[i])
        if i == end:                  # finished this level -> jump again
            jumps += 1
            end = farthest
    return jumps
```

Problem 2 — gas station. Stations on a circle; station i gives `gas[i]` and driving to the next costs `cost[i]`. Find the start index that completes the loop (−1 if none). Two facts: if total gas < total cost, no start works; and if the tank goes negative at station i after starting at s, then **no** station between s and i can work either: starting at one of them means arriving there with an empty tank, whereas the run from s arrived with a tank ≥ 0 and still failed at i. So jump the start to i + 1.

```python
def gas_station(gas, cost):
    if sum(gas) < sum(cost):
        return -1
    start = tank = 0
    for i, (g, c) in enumerate(zip(gas, cost)):
        tank += g - c
        if tank < 0:
            start, tank = i + 1, 0
    return start
```

Problem 3 — candy. Children stand in a line with ratings; each gets at least one candy and a child with a higher rating than a neighbour must get more than that neighbour. Minimise the total. One left-to-right pass satisfies the "left neighbour" rule, one right-to-left pass the "right neighbour" rule, and taking the `max` keeps both.

```python
def candy(ratings):
    n = len(ratings)
    c = [1] * n
    for i in range(1, n):
        if ratings[i] > ratings[i - 1]:
            c[i] = c[i - 1] + 1
    for i in range(n - 2, -1, -1):
        if ratings[i] > ratings[i + 1]:
            c[i] = max(c[i], c[i + 1] + 1)
    return sum(c)
```

Problem 4 — interval scheduling (`max_non_overlapping`): to fit the most non-overlapping talks in one room, sort by **end** time and take every talk that starts after the last one you took. The talk that ends first leaves the most room for the rest — that is the exchange argument. Sorting by start time or by length both fail on small counter-examples.

Output of `14_greedy.py`:

```
can_jump [3,2,1,0,4]: False | min_jumps [2,3,1,1,4]: 2
gas station start: 3
candy [1,3,4,5,2]: 11
max non-overlapping talks: 4
all asserts passed
```

Trade-offs

	Greedy	DP
Speed	usually O(n) or O(n log n)	states × work per state
Correctness	needs a proof; a plausible rule is often wrong	follows from the recurrence
Example of greedy failing	coins {1, 3, 4}, amount 6: greedy 4+1+1 = 3 coins; best 3+3 = 2	DP gets 2

Common mistakes

- Using a greedy rule without checking a small counter-example (the coin example above). If you cannot justify it in one sentence, consider DP.
- Candy: doing only one pass, or setting `c[i] = c[i+1] + 1` in the second pass and breaking the first pass's result (use `max`).
- Min jumps: looping to the last index inclusive, which counts one extra jump.

In the interview

30-second answer: "Greedy takes the locally best choice and never revisits it, so it's usually one pass after a sort. I only use it when I can justify it — typically an exchange argument, like 'the interval that ends first never hurts'. Otherwise I fall back to DP. Jump game tracks the farthest reachable index; gas station resets the start when the tank goes negative; candy is two passes."

- *How do you know greedy is right?* → Exchange argument, or "greedy stays ahead": after every step, greedy's partial answer is at least as good as any other.
- *Jump game with minimum jumps — why is it O(n) and not BFS O(n²)?* → Each level is a contiguous range, so you track only its end and the farthest reach.

19.15 Intervals

What it is

An **interval** is a pair `[start, end]` — a meeting, a booking, a range of IDs. Interval problems ask whether they overlap, how to merge them, or how many are active at once. Two intervals `[a, b]` and `[c, d]` **overlap** when `max(a, c) <= min(b, d)` (use `<` instead of `<=` if touching ends do not count — ask).

The problem it solves

Comparing every pair is O(n²). **Sorting** puts overlapping intervals next to each other, so most problems become one O(n) pass after an O(n log n) sort. Which key to sort by matters: **start** time for merging, **end** time for "keep as many as possible" (19.14).

How it works

Problem 1 — merge. Sort by start; each interval either overlaps the last kept one (extend its end) or starts a new one. (This is the snippet from 19.6.)

Problem 2 — insert into a sorted, non-overlapping list. No re-sort needed: copy the intervals that end before the new one starts, absorb every interval that overlaps it, then copy the rest. O(n).

```
def insert(intervals, new):
    out, i, n = [], 0, len(intervals)
    s, e = new
    while i < n and intervals[i][1] < s:          # entirely before
        out.append(intervals[i]); i += 1
    while i < n and intervals[i][0] <= e:         # overlapping -> absorb
        s, e = min(s, intervals[i][0]), max(e, intervals[i][1]); i += 1
    out.append([s, e])
    out.extend(intervals[i:])                     # entirely after
    return out
```

Problem 3 — intersection of two sorted lists. Two pointers (19.2): the overlap of the current pair is `[max(starts), min(ends)]` if non-empty; then advance whichever interval **ends first**, because it cannot overlap anything later in the other list.

Problem 4 — minimum meeting rooms. How many meetings run at the same time at the peak? Two standard answers:

```
def min_rooms_heap(meetings):          # heap of end times of meetings in progress
    ends = []
    for start, end in sorted(meetings):
        if ends and ends[0] <= start:  # the earliest-ending room is free again
            heapq.heapreplace(ends, end)
        else:
            heapq.heappush(ends, end)
    return len(ends)

def min_rooms_sweep(meetings):         # sweep line: +1 at a start, -1 at an end
    events = sorted([(s, 1) for s, _ in meetings] + [(e, -1) for _, e in meetings])
    return max(accumulate(d for _, d in events), default=0)
```

The **sweep line** idea: turn each interval into two events and walk them in time order, keeping a running count. Sorting the tuples puts `(t, -1)` before `(t, 1)`, so a meeting ending at 10 frees its room for one starting at 10.

Output of `15_intervals.py`:

```
merged: [[1, 6], [8, 10], [15, 18]]
insert [4,8]: [[1, 2], [3, 10], [12, 16]]
intersection: [[1, 2], [5, 5], [8, 10], [15, 23], [24, 24], [25, 25]]
rooms needed: 2
all asserts passed
```

The script also has `erase_overlaps` (fewest intervals to delete so the rest don't overlap = n − the greedy maximum from 19.14).

Trade-offs

Question	Sort by	Technique	Cost
merge / union	start	extend last kept	O(n log n)
insert one	already sorted	three phases	O(n)
overlap of two lists	already sorted	two pointers	O(m + n)
max active at once	start (heap) or event time (sweep)	min-heap of ends / running sum	O(n log n)
keep the most / delete the fewest	**end**	greedy	O(n log n)

Common mistakes

- Not asking whether `[1, 4]` and `[4, 5]` overlap; the `<=` vs `<` choice changes the answer.
- Merge: `out[-1][1] = end` instead of `max(out[-1][1], end)` — breaks when an interval is fully inside the previous one (`[1, 10]`, `[2, 3]`).
- Sweep line: letting starts sort before ends at the same time, which over-counts rooms.

In the interview

> **30-second answer:** "I sort the intervals — by start to merge or count overlaps, by end to keep as many as possible — and then make one pass. Two intervals overlap when max of starts ≤ min of ends. For meeting rooms I use a min-heap of end times or a sweep line of +1/−1 events; both are O(n log n)."

- *Which room does each meeting get?* → Keep `(end, room_id)` in the heap; reuse the popped room's id.
- *Intervals arrive as a stream?* → Keep them in a sorted structure keyed by start (`sortedcontainers.SortedList` or `bisect` on a list) and merge with neighbours on insert.

19.16 Fast & slow pointers

What it is

Two pointers walk the **same** sequence at different speeds: `slow` moves one step, `fast` moves two. This is **Floyd's cycle-finding algorithm** (also called "tortoise and hare"). It works on anything where "next" is defined — a linked list, or an implicit sequence like x → f(x) — using O(1) extra memory. 19.7 introduced it for the middle and a yes/no cycle check; here are the full set of uses.

The problem it solves

Detecting a loop or finding a midpoint usually needs a `set` of visited nodes (O(n) memory) or a first pass to count the length. Fast/slow does both in one pass with two variables.

How it works

- **Middle:** when `fast` reaches the end, `slow` has done half the steps. For an even length this returns the second middle.
- **Cycle start:** if there is a loop, `fast` gains one step per move on `slow` inside it, so they must meet. Then put one pointer back at the head and move both one step at a time: they meet exactly at the loop's entry. (Why: let a = distance from head to the entry, and let the meeting point be b steps into a loop of length c. `fast` walked twice as far as `slow`, which gives a = k·c − b for

some whole number k. So a steps from the head and a steps from the meeting point land on the same node, the entry.)

```
def cycle_start(head):
    slow = fast = head
    while fast and fast.next:
        slow, fast = slow.next, fast.next.next
        if slow is fast:                    # `is`: same node, not equal values
            slow = head
            while slow is not fast:
                slow, fast = slow.next, fast.next
            return slow
    return None
```

Problem 1 — palindrome linked list in O(1) space. Find the middle, reverse the second half in place (19.7's reversal), compare the halves, then reverse it back so the caller's list is unchanged.

Problem 2 — happy number. Replace n by the sum of the squares of its digits (19 → $1^2 + 9^2$ = 82 → 68 → 100 → 1). A number is **happy** if this reaches 1. The sequence always either hits 1 or loops forever, so treat `f(n)` as "next" and run Floyd:

```
def is_happy(n):
    def f(x):
        return sum(int(d) ** 2 for d in str(x))
    slow, fast = n, f(n)
    while fast != 1 and slow != fast:
        slow, fast = f(slow), f(f(fast))
    return fast == 1
```

Problem 3 — find the duplicate number. An array of n + 1 values, each in 1..n, one repeated, read-only, O(1) space. Treat index i → `nums[i]` as a "next" pointer: two indices point to the same value, so that value is the entry of a cycle — `cycle_start` on an array (`find_duplicate` in the script).

Output of `16_fast_slow_pointers.py`:

```
cycle starts at value: 2
palindrome [1,2,3,2,1]: True
happy numbers <= 30: [1, 7, 10, 13, 19, 23, 28]
all asserts passed
```

Complexity

All of these are O(n) time and O(1) extra space (n = list length, or the number of distinct values before the sequence repeats).

Common mistakes

- Loop condition `while fast.next and fast` — checks `fast.next` before knowing `fast` isn't `None`. Always `while fast and fast.next`.
- Comparing nodes with `==` on a `@dataclass` node: dataclasses generate a value-based `__eq__`, so two different nodes with equal fields compare equal (and on a cyclic list the comparison may never finish). Use `is`, or declare the node with `@dataclass(eq=False)` as the script does.
- Palindrome check that leaves the list half-reversed — interviewers ask "did you modify the input?"

In the interview

> **30-second answer:** "Slow moves one, fast moves two. When fast ends, slow is at the middle; if they meet, there's a cycle, and resetting one to the head and stepping both by one finds the entry. It's O(n) time and O(1) space, and it also works on implicit sequences — happy number, or find-the-duplicate where index → value is the next pointer."

- *Cycle length?* → After they meet, keep one still and count steps until the other returns.
- *Why not a set of visited nodes?* → Also O(n) time and simpler, but O(n) memory; mention it as the first solution, then optimise.

19.17 Sort & search: quicksort, quickselect, Dutch flag, merge sort on a list

What it is

You will normally call `sorted` (19.6), but interviewers still ask for the ideas inside sorting algorithms because they solve other problems:

- **Partition** — rearrange an array around a **pivot** value so smaller items come first and larger items after. It is the core of **quicksort** (partition, then sort both sides) and **quickselect** (partition, then continue into **one** side to find the k-th element).
- **Three-way partition** (Dutch national flag) — sort an array of only three distinct values in one pass.
- **Merge sort** — split in half, sort each half, merge. On a linked list it needs no extra arrays, which is why it is the list sorting algorithm.

The problem it solves

"Find the k-th largest" by sorting costs O(n log n); a heap (19.5) costs O(n log k); quickselect costs **O(n) on average** because each partition throws away one side. "Sort 0s, 1s and 2s" by counting takes two passes; the Dutch flag takes one.

How it works

Lomuto partition with a random pivot (a random pivot makes the $O(n^2)$ worst case — always picking the smallest item — extremely unlikely):

```python
def partition(nums, lo, hi, rng):
    p = rng.randint(lo, hi)
    nums[p], nums[hi] = nums[hi], nums[p]    # move the pivot to the end
    pivot, store = nums[hi], lo
    for i in range(lo, hi):
        if nums[i] < pivot:                  # grow the "smaller" region [lo, store)
            nums[i], nums[store] = nums[store], nums[i]
            store += 1
    nums[store], nums[hi] = nums[hi], nums[store]
    return store                             # pivot is now in its final sorted place

def kth_largest(nums, k, seed=0):
    nums, rng = nums[:], random.Random(seed)
    target = len(nums) - k                   # k-th largest = index n-k when sorted
    lo, hi = 0, len(nums) - 1
    while True:
        p = partition(nums, lo, hi, rng)
        if p == target:
            return nums[p]
        if p < target:
            lo = p + 1
        else:
            hi = p - 1
```

`quicksort` in the script uses the same `partition` with an explicit stack of `(lo, hi)` ranges instead of recursion.

Dutch national flag — three regions grow while an "unknown" region shrinks. Invariant: `[0, low)` are 0s, `[low, mid)` are 1s, `(high, end]` are 2s, `[mid, high]` not yet seen.

```python
def sort_colors(nums):
    low, mid, high = 0, 0, len(nums) - 1
    while mid <= high:
        if nums[mid] == 0:
            nums[low], nums[mid] = nums[mid], nums[low]
            low += 1; mid += 1
        elif nums[mid] == 1:
            mid += 1
        else:                                # a 2 goes to the back; the value swapped
            nums[mid], nums[high] = nums[high], nums[mid]   # in is unseen, so mid stays
            high -= 1
```

Merge sort on a linked list — find the middle with fast/slow pointers (19.16; start `fast` at `head.next` so `slow` stops at the end of the left half), cut, sort both halves recursively, then merge with a dummy head (19.7). `sort_list` in the script.

Output of `17_sort_search.py` (the script also checks `quicksort` and `kth_largest` against `sorted` on 200 random lists):

```
merge-sorted list: [1, 2, 2, 3, 4, 5]
quicksort: [1, 2, 4, 7, 8, 9] | 2nd largest of [3,2,1,5,6,4]: 5
sort colors: [0, 0, 1, 1, 2, 2]
all asserts passed
```

Trade-offs

Algorithm	Time (avg / worst)	Extra space	Stable?	Use when
`sorted` (Timsort)	O(n log n) / O(n log n)	O(n)	yes	always, in real code
quicksort	O(n log n) / O(n²)	O(log n) stack	no	asked to implement in place
merge sort	O(n log n) / O(n log n)	O(n) array, O(log n) list	yes	linked lists; external sorting (sorting data too big for memory, in chunks on disk) of huge files
quickselect	O(n) / O(n²)	O(1)	—	one k-th element, input may be reordered
heap top-k	O(n log k)	O(k)	—	streaming data, or k ≪ n
Dutch flag	O(n)	O(1)	no	exactly three key values

Common mistakes

- Always choosing the first or last element as pivot — O(n²) on sorted input. Randomise.
- Many equal values: Lomuto puts all equals on one side and degrades; a three-way partition (Dutch flag around the pivot) fixes it.
- Dutch flag: advancing `mid` after swapping with `high`.
- Quickselect mutating the caller's list without saying so — copy, or ask.

In the interview

30-second answer: "Partition puts a pivot in its final place with smaller items left and larger right. Quicksort recurses on both sides — O(n log n) average; quickselect only follows the side containing index n−k — O(n) average; random pivots avoid the worst case. Three distinct values: Dutch flag, one pass with low/mid/high. Linked lists: merge sort, splitting with fast/slow pointers."

- *Guaranteed O(n) selection?* → Median of medians (a pivot-choosing scheme that guarantees a reasonably central pivot), O(n) worst case but slower in practice; mention it, don't code it.
- *Quickselect vs heap for top-k?* → Quickselect is O(n) average but needs all data in memory and reorders it; a size-k heap works on a stream in O(k) memory.

19.18 Math & geometry

What it is

A family of problems where the trick is a small mathematical fact rather than a data structure: walking a matrix in a pattern, handling integer limits, comparing slopes exactly, or replacing a simulation with a formula. 19.12 has the number-theory toolkit (`gcd`, `pow`, `isqrt`, `comb`); this section applies it.

How it works

Problem 1 — spiral order. Keep four boundaries — `top`, `bottom`, `left`, `right` — and peel one ring at a time: along the top row, down the right column, back along the bottom row, up the left column, shrinking the matching boundary after each side. The two `if` checks stop a single remaining row or column from being read twice.

```python
def spiral_order(matrix):
    out = []
    if not matrix:
        return out
    top, bottom, left, right = 0, len(matrix) - 1, 0, len(matrix[0]) - 1
    while top <= bottom and left <= right:
        out += [matrix[top][c] for c in range(left, right + 1)];  top += 1
        out += [matrix[r][right] for r in range(top, bottom + 1)]; right -= 1
        if top <= bottom:
            out += [matrix[bottom][c] for c in range(right, left - 1, -1)]; bottom -= 1
        if left <= right:
            out += [matrix[r][left] for r in range(bottom, top - 1, -1)]; left += 1
    return out
```

`rotate` in the script turns an n × n matrix 90° clockwise in place: **transpose** (swap `m[i][j]` and `m[j][i]`), then reverse each row.

Problem 2 — reverse an integer with 32-bit overflow. Reverse the digits of x, but return 0 if the result falls outside the signed 32-bit range $[-2^{31}, 2^{31} - 1]$. Python ints never overflow (19.12), so you must check the range yourself. Interviewers often want the check a 32-bit language needs — test **before** multiplying: `res > (INT_MAX - digit) // 10`.

```python
INT_MIN, INT_MAX = -2**31, 2**31 - 1

def reverse_int(x):
    sign = -1 if x < 0 else 1
    x, res = abs(x), 0
    while x:
        x, digit = divmod(x, 10)
        if res > (INT_MAX - digit) // 10:
            return 0
        res = res * 10 + digit
    res *= sign
    return res if INT_MIN <= res <= INT_MAX else 0
```

(Taking `abs(x)` first avoids a trap: `-123 % 10` is 7 in Python, not −3, because `%` follows the sign of the divisor — see 19.19.)

Problem 3 — most points on one line (collinear points). For each anchor point, group the other points by the slope of the line through the anchor; the biggest group, plus the anchor and any exact duplicates of it, is the best line through that anchor. Store the slope as a **reduced fraction** `(dx // g, dy // g)` with `g = gcd(dx, dy)` and one fixed sign, not as the float `dy / dx` — floats can round two different slopes to the same value, and vertical lines would divide by zero.

Problem 4 — Josephus. n people stand in a circle; counting round, every k-th person leaves; who survives? Simulating with `list.pop` is $O(n^2)$. The recurrence $J(1) = 0$, $J(m) = (J(m-1) + k) \bmod m$ (0-based) gives it in O(n): after the first removal the circle is a problem of size m − 1 whose numbering is shifted by k.

```python
def josephus(n, k):
    survivor = 0
    for m in range(2, n + 1):
        survivor = (survivor + k) % m
    return survivor
```

Output of `18_math_geometry.py` (the script checks `josephus` against the simulation for every n < 30, k < 8):

```
spiral: [1, 2, 3, 4, 8, 12, 11, 10, 9, 5, 6, 7]
rotated: [[7, 4, 1], [8, 5, 2], [9, 6, 3]]
reverse 1534236469 -> 0 (overflow)
max collinear: 4
Josephus n=7, k=3 survivor index: 3
all asserts passed
```

Complexity

Problem	Time	Space
spiral / rotate	O(r·c)	O(1) extra (rotate in place)
reverse integer	O(number of digits)	O(1)
max points on a line	O(n²) (the gcd adds a log factor)	O(n) per anchor
Josephus	O(n)	O(1)

Common mistakes

- Spiral: missing the two `if` guards, so a middle row or column is output twice on non-square matrices.
- Slopes as floats, or not normalising the sign — `(1, -2)` and `(-1, 2)` are the same direction.
- Forgetting duplicates of the anchor point in "max points on a line".
- Relying on Python's big ints when the problem says "assume a 32-bit environment".

In the interview

30-second answer: "For matrix walks I keep explicit boundaries and shrink them; rotation is transpose plus reverse rows. Python ints don't overflow, so for 32-bit problems I check the range explicitly — before multiplying, as a C or Java solution would. For geometry I compare slopes as reduced integer fractions via `gcd`, never floats. When a simulation is slow, I look for a recurrence, like Josephus's `(J + k) % m`."

- *Rotate counter-clockwise?* → Transpose, then reverse each column (or reverse each row first, then transpose).
- *Why does Josephus's recurrence work?* → After removing person k−1, renumber the remaining m−1 people starting from the next one; the survivor of that smaller circle, shifted back by k (mod m), is the survivor of the larger one.

19.19 Common pitfalls

Each of these is demonstrated (bug, then fix) in `19_pitfalls.py`. Chapter 20 covers general Python gotchas in more depth; these are the ones that bite in timed coding.

1. [1, 2] [1, 2] <- shared list
   fixed: [1] [2]
2. [[0]*3]*2 -> [[9, 0, 0], [9, 0, 0]] | comprehension -> [[9, 0, 0], [0, 0, 0]]
3. shallow sees change: 99 | deep does not: 1
4. stored references: [[1, 2], [1, 2]] | use path[:] -> [[1], [1, 2]]
5. default limit: 1000
   depth(3000) -> RecursionError; use an explicit stack or raise the limit
6. 7 // 2 = 3 | -7 // 2 = -4 | int(-7 / 2) = -3 | -7 % 3 = 2
   7 / 2 = 3.5 (always float) | round(2.5) = 2 | round(3.5) = 4
7. 0.1 + 0.2 == 0.3: False | isclose: True
8. a == b: True | a is b: False | use `is` only for None/sentinels
9. remove while iterating -> [1, 2, 3] | comprehension -> [1, 3]
10. ''.join(parts) = 01234 (O(n); repeated += can be O(n^2))
11. xs.sort() returns None - xs is now [1, 2, 3] | sorted() returns a new list
12. list still callable: ['a', 'b'] [1, 2]

#	Pitfall	Why	Fix
1	Mutable default `def f(x, acc=[])`	defaults are evaluated **once** when `def` runs; every call shares the list	`acc=None`, then `if acc is None: acc = []`
2	`[[0] * n] * m`	the outer `*` repeats a **reference** to one row	`[[0] * n for _ in range(m)]`
3	Shallow copy of nested data	`a[:]`, `list(a)`, `a.copy()`, `copy.copy` copy only the outer list	`copy.deepcopy`, or `[row[:] for row in grid]` for 2-D
4	`res.append(path)` in backtracking	stores the same list object	`res.append(path[:])`
5	Deep recursion	limit 1000, no tail-call optimisation	iterative stack/deque; `sys.setrecursionlimit` as last resort
6	Integer division	`//` **floors** (toward −∞); `int(a / b)` truncates toward 0; `%` takes the sign of the divisor; `/` always returns float; `round` uses banker's rounding (ties to even)	choose deliberately; for "truncate toward zero" use `int(a / b)` (small ints) or sign-aware `//`
7	Float equality	binary floats can't represent 0.1 exactly	`math.isclose`, or stay in integers
8	`is` vs `==`	`is` compares identity	`==` for values; `is` only for `None` and sentinels
9	Mutating a list while iterating	the iterator's index skips elements	build a new list, or iterate over a copy
10	String `+=` in loops	may copy each time → O(n²)	collect parts, `"".join`
11	`nums = nums.sort()`	in-place methods return `None`	`nums.sort()` alone or `nums = sorted(nums)`
12	Shadowing built-ins (`list`, `dict`, `sum`, `max`, `input`, `id`)	later calls hit your variable	pick other names (`nums`, `total`, `list_`)

Two more to remember: `float("inf")` compares fine with ints but `int(float("inf"))` raises `OverflowError`; and `@cache` on a helper with global state gives stale answers across test cases (19.10).

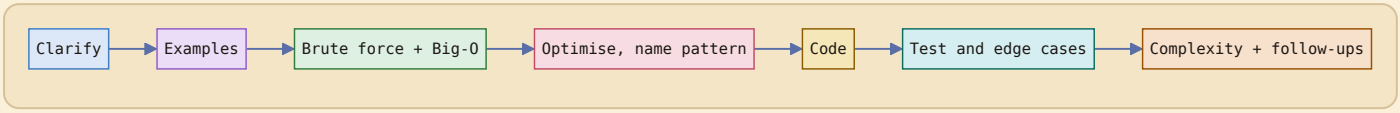
19.20 How to talk through a solution

What it is

Interviewers score **communication and process** as well as the final code; a predictable structure keeps you calm and shows how you think.

How it works — the seven steps

1. **Clarify** (1–2 min): input sizes, ranges (negatives? duplicates? empty?), output format, may you modify the input? Size hints the target: $n \le 20$ → exponential is fine; $n \le 10^3$ → $O(n^2)$; $n \le 10^5$–10^6 → $O(n \log n)$ or $O(n)$.
2. **Examples:** one normal case and one edge case by hand.
3. **Brute force:** state it and its complexity in one sentence — proof you understand the problem, and a fallback.
4. **Optimise:** name the pattern ("sorted → two pointers", "k-th → heap"), explain *why* it's faster, agree before coding.
5. **Code:** clear names, small helpers, idioms (`enumerate`, `defaultdict`, unpacking); narrate briefly.
6. **Test:** trace your example, then edge cases — empty, single element, all equal, negatives, maximum sizes.
7. **Analyse:** time and space, and what changes for follow-ups (streaming data, memory limits).



Phrases that help

- "Before I optimise, the brute force is $O(n^2)$ because we check every pair."
- "Since the array is sorted, I can discard half each step — binary search, $O(\log n)$."
- "I'll use a `deque` here rather than a list because `popleft` is $O(1)$."
- "Space is $O(n)$ for the hash map; if memory mattered, I could sort and use two pointers at $O(1)$ extra space but $O(n \log n)$ time."

Python-specific tips

- Using `Counter`, `heapq`, `bisect`, `itertools` is expected — but know each call's complexity and be ready to hand-write it ("could you do this without `Counter`?").
- Type hints in the signature (`def solve(nums: list[int]) -> int:`) read well; don't golf ("code golf" = squeezing code into as few characters as possible) — a clear four-line loop beats a one-liner you can't explain.
- Know which Python version the platform runs; newer helpers (`itertools.pairwise`/`math.lcm` need 3.9–3.10, `itertools.batched` 3.12, `heapq.heappush_max` 3.14) may not exist there.

Chapter TL;DR

Pattern	Reach for it when…	Python tool
Costs	always state them	list = array (O(1) end, O(n) front/search); dict/set O(1) avg; slices and `+` copy
Two pointers / window / prefix sums	sorted pairs; contiguous + monotonic rule; range sums or "sum = k"	indices, `accumulate`, dict of prefixes
Hashing	seen-before, counts, groups	`set`, `dict`, `Counter`, `defaultdict` (hashable keys)
Stack / queue / monotonic	brackets, "next greater", window max	`list`, `deque`
Heap	top-k, merge-k, two heaps (running median, capital), Dijkstra	`heapq` (negate for max, tie-breakers)
Sort / binary search	ordering, "first position", monotonic feasibility, weighted random	`key=` tuples, `bisect_left/right`, prefix sums
Lists / trees / graphs	pointers, traversals, shortest paths, dependencies	node classes, `deque` BFS, `heapq` Dijkstra, Kahn, DSU
DP / backtracking	overlapping subproblems; enumerate with pruning	`@cache` or tables; choose → explore → un-choose
Bits / math	masks, parity, number theory	`x & (x−1)`, `pow(b, e, m)`, `gcd`, `isqrt`, `comb`
Tries	prefix search, autocomplete, wildcard, many words on a grid	dict-of-children nodes + end-of-word field
Greedy	a locally best choice is provably safe (exchange argument)	one pass, often after `sorted(key=...)`
Intervals	merge, insert, overlap, rooms needed	sort by start (merge) or end (keep most); heap of ends; sweep line
Fast & slow pointers	cycle and its entry, middle, O(1)-space list checks, implicit sequences	`slow`, `fast`; compare with `is`
Sort & search	k-th element, partition in place, three values, sort a linked list	quickselect, Dutch flag, merge sort
Math & geometry	matrix walks, 32-bit limits, collinear points, circle elimination	boundaries, explicit range checks, `gcd` slopes, recurrences

Avoid the pitfalls of 19.19 (mutable defaults, `[[0]*n]*m`, shallow copies, the 1000-frame recursion limit, `//` flooring), and narrate: clarify → brute force → optimise → code → test → analyse.